TASK 05-50-03-200-802-A
EFFECTIVITY: ALL

1. Hard Landing And Off Temperature Envelope Landing - Inspection/Check

A. General

(1) After an unusual landing condition, where a hard landing can occur, it is necessary to do inspections to guarantee the aircraft integrity.

(2) An unusual landing condition is considered when the loads are out of the design envelope. This envelope includes a combination of the aircraft weight, vertical CG acceleration, roll rate, pitch rate and ambient temperature at touchdown.

(3) This task includes:

(a) Hard Landing - it is an exceedance of the vertical CG acceleration, roll rate, or pitch rate thresholds at touchdown.

(b) Off-temperature landing - it is an exceedance of the temperature envelope at touchdown.

(4) If an overweight landing occurs, refer to AMM TASK 05-50-27-210-802-A/600.

(5) A hard landing inspection is necessary when the crew members report a possible hard landing based on their judgment; or when the analysis of the flight operation data shows that the envelope of the landing parameters was not obeyed.

(6) The flight data source can be either DVDR 1, DVDR 2, QAR. Any of them is sufficient to define if or not a hard landing has occurred.

NOTE: DVDR 1 is installed in the forward avionics compartment and DVDR 2 is installed in the aft avionics compartment.

(7) When the aircraft is equipped with the QAR, you can use this equipment as an alternative for the DVDR 2.

NOTE: • The QAR must be operating in continuous mode to be used as alternative to DVDR 2.

(8) The operator can do the analysis of the flight operation data, but if the alternative is to send the data for an EMBRAER analysis, it is recommended to send the QAR file.

(9) These maintenance technicians are necessary to do this task:

- 2 - Airframe (Structure)

B. References

REFERENCE	*DESIGNATION*
AMM SDS 52-11-00/1	FORWARD PASSENGER DOOR
AMM SDS 52-12-00/1	AFT PASSENGER DOOR
AMM SDS 52-31-00/1	FORWARD CARGO DOOR
AMM SDS 52-32-00/1	AFT CARGO DOOR

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REFERENCE	*DESIGNATION*
AMM SDS 52-41-00/1	FORWARD SERVICE DOOR
AMM SDS 52-42-00/1	AFT SERVICE DOOR
AMM TASK 05-50-27-210-802-A/600	Overweight Landing - Inspection/Check
AMM TASK 06-41-00-800-801-A/100	Fuselage Access Doors and Panels
AMM TASK 07-10-01-500-801-A/200	Complete Aircraft Jacking - Maintenance Practices
AMM TASK 07-10-01-500-802-A/200	Complete Aircraft Lowering - Maintenance Practices
AMM TASK 20-00-00-910-801-A/200	Aircraft Maintenance Safety Procedures - Standard Procedures
AMM TASK 24-23-00-710-801-A/500	Ram-Air-Turbine (RAT) Manual Deployment - Operational Test
AMM TASK 24-42-02-860-801-A/200	Connection of the external AC power supply to the aircraft - Aircraft/System Configuration
AMM TASK 24-42-02-860-802-A/200	Disconnection of the external AC power supply from the aircraft - Aircraft/System Configuration
AMM TASK 31-31-00-970-801-A/200	DVDR System - Download Procedure with PCMCIA Card
AMM TASK 31-31-00-970-802-A/200	DVDR System - Download Procedure with Portable Unit
AMM TASK 31-31-00-970-804-A/200	DVDR System - Download Procedure via Universal Software
AMM TASK 32-10-00-200-801-A/600	Main-Landing-Gear and Doors - Inspection
AMM TASK 32-11-01-000-801-A/400	Main-Landing-Gear Shock Strut - Removal
AMM TASK 32-11-03-200-802-A/600	Main-Landing-Gear Shimmy Damper - Inspection
AMM TASK 32-11-05-000-801-A/400	Main-Landing-Gear Side Stay - Removal
AMM TASK 32-11-05-400-801-A/400	Main-Landing-Gear Side Stay - Installation
AMM TASK 32-11-07-000-801-A/400	Main-Landing-Gear Locking Stay - Removal
AMM TASK 32-11-11-000-801-A/400	Main-Landing-Gear Torque Link -Removal
AMM TASK 32-20-00-200-801-A/600	Nose-Landing-Gear and Doors - Inspection
AMM TASK 32-21-03-000-801-A/400	Nose-Landing-Gear Drag Brace - Removal
AMM TASK 32-21-05-000-801-A/400	Nose-Landing-Gear Locking Stay - Removal
AMM TASK 32-49-01-000-801-A/400	Nose Wheel - Removal
AMM TASK 32-49-11-000-801-A/400	Brake Assembly - Removal
AMM TASK 32-49-11-400-801-A/400	Brake Assembly - Installation
AMM TASK 32-61-03-820-801-A/200	Nose-Landing-Gear Downlock Sensor (Clearance) - Adjustment
AMM TASK 32-61-07-820-801-A/200	Main-Landing-Gear Downlock Sensor (Clearance) - Adjustment
AMM TASK 32-62-01-820-801-A/200	Nose-Landing-Gear Weight-on-Wheels Sensor (Clearance) - Adjustment
AMM TASK 32-62-03-820-801-A/200	Main-Landing-Gear Weight-on-Wheels Sensor (Clearance) - Adjustment
AMM TASK 49-91-05-200-801-A/600	Magnetic Drain Plug- Inspection

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REFERENCE	*DESIGNATION*
AMM TASK 51-50-00-800-801-A/600	Aircraft Alignment Check
AMM TASK 53-04-20-000-801-A/400	Forward Wing-to-Fuselage Fairing - Removal
AMM TASK 53-04-20-400-801-A/400	Forward Wing-to-Fuselage Fairing - Installation
AMM TASK 53-04-40-000-801-A/400	Lateral Wing-to-Fuselage Fairing - Removal
AMM TASK 53-04-40-400-801-A/400	Lateral Wing-to-Fuselage Fairing - Installation
AMM TASK 53-04-50-000-801-A/400	Aft Wing-to-Fuselage Fairing - Removal
AMM TASK 53-04-50-400-801-A/400	Aft Wing-to-Fuselage Fairing - Installation
AMM TASK 57-51-00-200-801-A/600	Flap Rod Assy - Inspection
LMM TASK 72-00-00-220-801	OVERLIMIT CONDITION INSPECTION

C. Zones and Accesses

ZONES ACCESS	*LOCATION*	*REFERENCE*
125AZ	LH FWD Fuselage Bottom - Rear Area	AMM TASK 06-49-00-800-801-A/100
126AZ	RH FWD Fuselage Bottom - Rear Area	AMM TASK 06-49-00-800-801-A/100
191AR	Wing-to-Fuselage Fairing FWD Panel	AMM TASK 06-41-00-800-801-A/100
191BL	Wing-to-Fuselage Fairing FWD Panel	AMM TASK 06-41-00-800-801-A/100
191CL	Wing-to-Fuselage Fairing FWD Panel	AMM TASK 06-41-00-800-801-A/100
191EL	Wing-to-Fuselage Fairing FWD Panel	AMM TASK 06-41-00-800-801-A/100
192AL	Wing-to-Fuselage Fairing Center Panel	AMM TASK 06-41-00-800-801-A/100
192BL	Wing-to-Fuselage Fairing Center Panel	AMM TASK 06-41-00-800-801-A/100
192BR	Wing-to-Fuselage Fairing Center Panel	AMM TASK 06-41-00-800-801-A/100
192CR	Wing-to-Fuselage Fairing Center Panel	AMM TASK 06-41-00-800-801-A/100
195AL	Wing-to-Fuselage Fairing Aft Panel	AMM TASK 06-41-00-800-801-A/100
195AR	Wing-to-Fuselage Fairing Aft Panel	AMM TASK 06-41-00-800-801-A/100
195BL	Wing-to-Fuselage Fairing Aft Panel	AMM TASK 06-41-00-800-801-A/100
195CR	Wing-to-Fuselage Fairing Aft Panel	AMM TASK 06-41-00-800-801-A/100
195DL	Wing-to-Fuselage Fairing Aft Panel	AMM TASK 06-41-00-800-801-A/100

D. Hard-Landing Detection Procedure

SUBTASK 210-054-A

(1) Figure 601, Figure 602, Figure 603, Figure 604, and Figure 605 show flowcharts that summarize of the necessary actions when the crew or ACMF reports a possible hard landing, when the hard landing is detected by flight data analysis or when the aircraft landed out of the temperature envelope.

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(2) *EFFECTIVITY: ON AIRCRAFT WITH HONEYWELL DVDR*

When the crew or ACMF reports a possible hard landing follow the steps according to Figure 601, do one of the following options (a) or (b):

NOTE: ACMF reports for roll rate analysis have been found to be conservative. If the flight crew has not identified a hard landing, the ACMF report for roll rate exceedance may be disregarded due to the limitation of this application. The ACMF reports or the absence of these do not replace or have any priority over flight crew reports.

(a) Proceed to DVDR data analysis from figures Figure 602, Figure 603, and Figure 604, or;

(b) Do all phase-I inspection for both aircraft structure and engines before the next flight.

1 If no discrepancy is found at aircraft structure a flyby of 10 FC is allowed for it. As well, if no discrepancy is found for the engines a flyby of 10 FC is allowed for them.

2 During the flyby period, define if the flight data analysis to detect hard landings will be performed or not.

3 If flight analysis will be performed follow the procedures according to flowcharts from Figure 602, Figure 603, and Figure 604 described on the sections below.

NOTE: The flight data download must be done within a maximum of 20 FH. If not, the data can be missed due to the recording time of the unit.

4 If flight data analysis will not be performed, within the flyby period, do all phase-II inspections for the aircraft structure and/or engines, if any of them have not been performed yet.

a Make sure that phase-I inspections have been completed for both aircraft structure and engines before starting phase-II inspections.

b If any discrepancy is detected for the aircraft structure on phase-II inspection, proceed to phase-III inspection. Send flight data and data from all finding to Embraer for analysis and disposition. Repair the aircraft is applicable and put the aircraft back to service.

NOTE:
- Make sure to complete all phase-II inspections before starting phase-III.
- Whenever phase-III is required for aircraft structure, engines phase-II inspection is mandatory.

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c If any discrepancy is detected on engines phase-II inspection follow the instructions provided on GE Engine Manual overlimit condition inspection LMM TASK 72-00-00-220-801 (Subtask 72-00-00-280-002) regarding to engine serviceability thresholds and flyby periods. Send data to Engine manufacturer for analysis and disposition if necessary and repair engines if applicable. After that engines will be cleared for service.

(c) In case discrepancy is found on phase-I inspections either for the aircraft structure or for the engines proceed immediately to the applicable phase-II inspections.

1 Make sure that phase-I inspections have been completed before starting phase-II inspections.

2 If phase-II inspections are applicable for the aircraft structure and any discrepancy is detected proceed to phase-III inspection. Send flight data and data from all findings to Embraer for analysis and disposition. Repair the aircraft if applicable and put the aircraft back to service.

NOTE:
- Make sure to complete phase-II inspections before starting phase-III.
- Whenever phase-III is required for the aircraft structure, engines phase-II inspection is mandatory.

3 If phase-II inspection is applicable for the engines and any discrepancy is detected, follow the instructions provided on GE Engine Manual overlimit condition inspection LMM TASK 72-00-00-220-801 (Subtask 72-00-00-280-002) regarding to engine serviceability thresholds and flyby periods. Send data to Engine manufacturer for analysis and disposition if necessary and repair engines if applicable. After that engines will be cleared for service.

(3) *EFFECTIVITY: 170:00344-00344*

When the crew or ACMF reports a possible hard landing follow the steps according to Figure 601, do one of the following options (a) or (b):

NOTE: ACMF reports for roll rate analysis have been found to be conservative. If the flight crew has not identified a hard landing, the ACMF report for roll rate exceedance may be disregarded due to the limitation of this application. The ACMF reports or the absence of these do not replace or have any priority over flight crew reports.

(a) Proceed to DVDR data analysis from figures Figure 602, Figure 603, and Figure 604, or;

(b) Do all phase-I inspection for both aircraft structure and engines before the next flight.

1 If no discrepancy is found at aircraft structure a flyby of 10 FC is allowed for it. As well, if no discrepancy is found for the engines a flyby of 10 FC is allowed for them.

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2 During the flyby period, define if the flight data analysis to detect hard landings will be performed or not.

3 If flight analysis will be performed follow the procedures according to flowcharts from Figure 602, Figure 603, and Figure 604 described on the sections below.

NOTE: The flight data download must be done within a maximum of 120 FH. If not, the data can be missed due to the recording time of the unit.

4 If flight data analysis will not be performed, within the flyby period, do all phase-II inspections for the aircraft structure and/or engines, if any of them have not been performed yet.

a Make sure that phase-I inspections have been completed for both aircraft structure and engines before starting phase-II inspections.

b If any discrepancy is detected for the aircraft structure on phase-II inspection, proceed to phase-III inspection. Send flight data and data from all finding to Embraer for analysis and disposition. Repair the aircraft is applicable and put the aircraft back to service.

NOTE:
- Make sure to complete all phase-II inspections before starting phase-III.
- Whenever phase-III is required for aircraft structure, engines phase-II inspection is mandatory.

c If any discrepancy is detected on engines phase-II inspection follow the instructions provided on GE Engine Manual overlimit condition inspection LMM TASK 72-00-00-220-801 (Subtask 72-00-00-280-002) regarding to engine serviceability thresholds and flyby periods. Send data to Engine manufacturer for analysis and disposition if necessary and repair engines if applicable. After that engines will be cleared for service.

(c) In case discrepancy is found on phase-I inspections either for the aircraft structure or for the engines proceed immediately to the applicable phase-II inspections.

1 Make sure that phase-I inspections have been completed before starting phase-II inspections.

2 If phase-II inspections are applicable for the aircraft structure and any discrepancy is detected proceed to phase-III inspection. Send flight data and data from all findings to Embraer for analysis and disposition. Repair the aircraft if applicable and put the aircraft back to service.

NOTE:
- Make sure to complete phase-II inspections before starting phase-III.
- Whenever phase-III is required for the aircraft structure, engines phase-II inspection is mandatory.

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3 If phase-II inspection is applicable for the engines and any discrepancy is detected, follow the instructions provided on GE Engine Manual overlimit condition inspection LMM TASK 72-00-00-220-801 (Subtask 72-00-00-280-002) regarding to engine serviceability thresholds and flyby periods. Send data to Engine manufacturer for analysis and disposition if necessary and repair engines if applicable. After that engines will be cleared for service.

(4) When the source for you to find a hard-landing condition is the flight data analysis, not a crew or ACMF reports, you must do the required inspections according to Figure 602, Figure 603, and Figure 604. Those are flowcharts with procedures to define if and what inspections are necessary based on monitoring data recorded during touchdown events.

(5) Due to the complexity of landing loads and their effect on the aircraft structure, three separate monitors are used with different thresholds to evaluate the event severity. Also the procedures and procedures and thresholds for each monitor differ between aircraft structure and aircraft engines.

(a) Hard landing analysis based on vertical acceleration monitor (Figure 602) indicates events that occur when the aircraft hits hard the ground over the main landing gears. These events affect beyond the main landing gears, the wings, fuselage, tail, and big mass items such as engines, APU and electronic racks.

(b) Hard landing analysis based on roll rate monitor (Figure 603) indicates events that occur when the aircraft hits hard the ground over one main landing gear. These events affect normally one of the main landing gears, one side of the wing, and the engines.

1 The use of the roll rate monitor must be validated based on the level of vertical acceleration detected at the same event (Figure 614). This is needed once the high roll rates may occur with the aircraft still airborne.

(c) Hard landing analysis based on pitch rate monitor (Figure 604) indicates events that occur when the aircraft hits hard the ground over the nose landing gear after landing de-rotation. These events affect the nose landing gear, front fuselage, and front fuselage big mass items the random air turbine and electronic racks

1 No engines thresholds need to be verified on this monitor. However if an aircraft phase-III inspection is required on this analysis, engines phase-II inspection must be performed.

NOTE:
- In situations when the aircraft touches the ground with the NLG first, none of hard landing detection procedures using flight data are applicable and all phase-II inspection for the aircraft structure are required.
- In this case if a phase-III inspection is required for the aircraft structure, engines phase-II inspection is mandatory.

(6) If the landing gear that may have exceeded the envelope is known, use the applicable procedure as follows:

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(a) A hard landing over any main landing gear will require analysis and inspections indicated on Figure 602, and Figure 603.

(b) A hard landing over the nose landing gear will require analysis and inspections indicated on Figure 604.

NOTE:
- In situations when the aircraft touches the ground with the NLG first, none of hard landing detection procedures using flight data are applicable and all phase-II inspections for the aircraft structure are required.
- In this case if a phase-III inspection is required for the aircraft structure, engines phase-II inspections is mandatory.
- If the landing gear that may have exceeded the envelope is not know or there are doubts about the event do the complete analysis according to Figure 602, Figure 603, and Figure 604.

(7) When the crew members report a possible hard landing, do the complete analysis according to Figure 602, Figure 603, and Figure 604.

(8) Use the parameters below to make an analysis of the flight data to find a hard landing:

- Vertival CG acceleration (NormAccel)
- Pitch angle (Pitch)
- Air/Ground indication (Air/Gnd)
- Gross weight (GrossWeight)
- Roll (Roll)
- Time (GMT)
- Date (Day)
- Flight number (FltNumber)

NOTE: The data must be selected from flight data source at rate of 8 sps (samples per second). A higher rate if used will not be compatible with the monitors thresholds.

(9) The procedures herein given, let you find a hard-landing condition, find the required inspections, and put the related aircraft into a serviceable condition.

(10) If you will not do Phase-III inspection for the aircraft structure, it is not necessary to send data to Embraer. But, if you will, send the data to Embraer for analysis (flight data and a report of the damage(s) found).

(11) As well, if the engines are cleared for service based on GE Engine Manual overlimit condition inspection LMM TASK 72-00-00-220-801 (Subtask 72-00-00-280-002) it is not necessary to send data to Engine Manufacturer. If not, send data to Engine Manufacturer for analysis (report of the damage (s) found).

(12) When the conditional inspection shows that you must examine a component, look for the conditions that follow:

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- Pulled-apart structure
- Loose paint (paint flakes)
- Discoloration
- Distortion (twisted parts)
- Wrinkles or buckles in the structure
- Bent components
- Fastener holes that become oversized
- Loose fasteners
- Fasteners that pulled out or are missing
- Delamination
- Fiber breakouts
- Black signs around fasteners in composites ("smoking" fasteners)
- Misalignment
- Interference
- Nicks or gouges
- Scratches
- Other signs of damage

(13) All the inspections are classified as GVI.

E. Hard-Landing Detection based on Off temperature landing

SUBTASK 200-006-A

(1) If a landing was done out of the temperature envelope (above ISA+35°C or below -45°C), none of the hard landing detection procedures using flight data are applicable. Then, do as follows:

(a) If the landing was done above ISA+35°C, contact Embraer for analysis and disposition.

(b) If the landing was done below -45°C, do all phase-II inspection for both aircraft structure and engines.

1 Do phase-III inspection only if you find a discrepancy during the phase-II inspections. Send flight data and data from all findings to Embraer for analysis and disposition. Repair the aircraft if applicable and put the aircraft back to service.

NOTE: • Make sure to complete all phase-II inspections before starting phase-III.

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2 If any discrepancy is detected on engines phase-II inspection follow the instructions provided on GE Engine Manual overlimit condition inspection LMM TASK 72-00-00-220-801 (Subtask 72-00-00-280-002) regarding to engine serviceability thresholds and flyby periods. Send data to Engine Manufacturer for analysis and disposition if necessary and repair the engines if applicable. After that engines will be cleared for service.

(c) If the landing was done above -45°C and there are was a hard landing report, follow the procedures for hard landing analysis (Figure 601 and/or Figure 602, Figure 603, and Figure 604).

F. Hard-Landing detection based on vertical acceleration monitor and roll rate monitor

SUBTASK 210-038-A

(1) Hard landing detection procedures based on vertical acceleration monitor and roll rate monitor are applicable when the aircraft hits hard the ground over any of main landing gears.

(2) The excess of one or the two parameter thresholds will classify the landing as hard landing.

NOTE:
- The use of the roll rate monitor must be validated based on the level of vertical acceleration detected at the same event (Figure 614).
- On the hard landing analysis if the landing gear that may have exceeded the envelope is not know or there are doubts about the event, make sure to do the analyses based on all hard landing monitors.
- When the crew members report a possible hard landing and the flight data will be analyzed, make sure to do analyses based on all hard landing monitors.

(3) *EFFECTIVITY: ON ACFT WITH HONEYWELL DVDR OR PRE-MOD SB 170-31-0050/10*

To do this detection it is necessary to download the flight data from DVDR 1 or DVDR 2 or QAR. Refer to AMM TASK 31-31-00-970-801-A/200 or AMM TASK 31-31-00-970-802-A/200 or , as necessary.

(4) Vertical Acceleration Monitor

(a) The steps described below for hard landing analysis based on the vertical acceleration monitor are summarized on the flowchart from (Figure 602).

(b) In the (downloaded) Flight Data, mark the initial and the final data samples to specify the range for the vertical-CG-acceleration analysis (Figure 606):

1 The range starts 4 seconds before the first "Ground" recorded for the air/ground indication (Air/Gnd).

2 The range ends at the first pitch angle (Pitch) sample with a value of less than or equal to 1.0 degrees after the transition of the Air/Ground indication (Air/Gnd) from "Air" to "Ground".

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(c) Get the maximum value of vertical CG acceleration (NormAccel) in the selected range.

NOTE:
- If the maximum value is between X.YZ00 and X.YZ49, it should be rounded down to X.(YZ);
- If the maximum value is between X.YZ50 and X.YZ99, it should be rounded up to X.(YZ+1).

(d) Get the gross weight (GrossWeight) recorded by the Flight Data Source during the air-to-ground transition.

(e) With these two parameters (maximum vertical CG acceleration and gross weight):

- Use flowchat of Figure 607 to determine the actions to be taken:

1 - Aircraft structure

- If the vertical CG acceleration peak recorded by the Flight Data Source is less than or equal to the low threshold for the aircraft gross weight, the aircraft structure is cleared for service based on the vertical acceleration monitor.
- If the vertical CG acceleration peak recorded by Flight Data Source is between the low threshold and the high load threshold for the aircraft gross weight, do the phase-I and phase-II inspection indicated for the aircraft structure when the vertical acceleration monitor is exceeded, at a scheduled maintenance opportunity (not exceeding 6,250 FC or 7,500 FH) or when 50 hard landing events occur, whichever occurs first.
- If the vertical CG acceleration peak recorded by the Flight Data Source is greater than the high load threshold for the aircraft gross weight, do the

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Phase-II inspection indicated for the aircraft structure when the vertical acceleration monitor is exceeded.

NOTE:
- Make sure that the Phase-I inspection for the aircraft structure is performed before starting the Phase-II.
- If it is not possible to do Phase-II inspection immediately, refer to Figure 615 for fly-by instructions. The fly-by is granted from the moment of the hard landing detection through the flight data analysis, provided that the phase-I inspection is performed immediately and no damage is found during this inspection.
- If a discrepancy caused by the hard landing issue is found during the Phase-I inspection, the Phase-II inspection must be done immediately.
- During the fly-by period, the Phase-II inspection must be done.

2 - Engines

- If the vertical CG acceleration peak recorded by the flight data source is less than or equal to the engine threshold, the engines are cleared for service based on the vertical acceleration monitor.
- If the vertical CG acceleration peak recorded by the flight data source is greater than the engine threshold, do the phase-II inspection indicated for the engine according to GE Engine Manual overlimit condition inspection LMM TASK 72-00-00-220-801 (Subtask 72-00-00-280-002).

NOTE:
- Make sure that phase-I inspection for the engine is performed before starting the phase-II inspection.
- If it is not possible to do phase-II inspection immediately, refer to Figure 615 for fly-by instructions. The fly-by is granted from the moment of the hard landing detection through the flight data analysis, provided that the phase-I inspection is performed immediately and no damage is found during this inspection.
- If a discrepancy caused by the hard landing issue is found during the Phase-I inspection, the Phase-II inspection must be done immediately.
- During the fly-by period, the Phase-II inspection must be done.

(f) If no discrepancy is found during the Phase-II inspections for the aircraft structure, it is cleared for serviced. But, if you find any discrepancy, proceed as follows:

1 Do the aircraft structure phase-III inspections and send Embraer these data;

a Flight data source that contain the hard landing incident;

b Damage report with information about the damage found during the inspections.

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2 Wait for Embraer disposition about the damage found and repair instructions before returning the aircraft to service.

3 Repair the aircraft structure if applicable.

4 After that, the aircraft structure is cleared for service.

(g) For the engines, if no serviceability threshold was exceeded according to GE manual, they are cleared for service. But, if you find any discrepancy, proceed as follows:

1 Check for fly-by instructions on engine manual and send findings to Engine Manufacturer.

2 If there is a fly-by granted engines are serviceable for the fly-by period.

3 Wait for engine manufacturer disposition about the damage found and repair instructions before returning the engines to service.

4 Repair the engines if applicable.

5 After that, the engines is cleared for service.

(5) Roll Rate Monitor

(a) The steps described below for hard landing analysis based on the roll rate monitor are summarized on the flowchart from Figure 603.

(b) Before starting the roll rate analysis, check if the maximum vertical CG acceleration peak recorded by flight data source exceeds the threshold from Figure 614 for the aircraft gross weight on suspect landing event. If not, skip the roll rate analysis.

(c) In the flight data, mark the initial and the final data samples to specify the range for roll data for analysis (Figure 606):

1 The range starts 2 seconds before the highest value recorded for the vertical CG acceleration.

2 The range stops 2 seconds after the highest value recorded for the vertical CG acceleration

(d) With the selected range of roll (Roll) values, calculate the roll rate. Refer to Figure 608.

(e) Get the gross weight parameter (GrossWeight) recorded by the DVDR during the air-to-ground transition.

(f) With these two parameters (maximum roll rate and gross weight), use the chart in Figure 608 to determine the actions to be taken:

- 1– Aircraft Structure:

- If the maximum roll rate recorded by the flight data source is less that or equal to the low threshold for the aircraft gross weight, the aircraft structure is cleared for service based on the roll rate monitor.

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- If the maximum roll rate recorded by Flight Data Source is greater than the threshold for the aircraft gross weight, do the phase-II inspection indicated for the aircraft structure when the roll rate monitor is exceeded.

NOTE:
- Make sure that the Phase-I inspection for aircraft structure is performed before starting the Phase-II.
- If it is not possible to do Phase-II inspection immediately, refer to Figure 616 for fly-by instruction. The fly-by is granted from the moment of the hard landing detection through the flight data analysis, provided that the Phase-I inspection is performed immediately and no damage is found during this inspection.
- If a discrepancy caused by the hard landing issue is found during the Phase-I inspection, the Phase-II inspection must be done immediately.
- During the fly-by period, the Phase-II inspection must be done.
- 2– Engines:
- If the maximum roll rate recorded by Flight Data Source is less than or equal to the engine threshold, the engines are cleared for service based on the roll rate monitor.
- If the maximum roll rate recorded by Flight Data Source is greater than the engine threshold, do the phase-II inspection indicated for the engine according to GE Engine Manual overlimit condition inspection LMM TASK 72-00-00-220-801 (Subtask 72-00-00-280-002).

NOTE:
- Make sure that the Phase-I inspection for the engines is performed before starting the Phase-II.
- If it is not possible to do Phase-II inspection immediately, refer to Figure 616 for fly-by instruction. The fly-by is granted from the moment of the hard landing detection through the flight data analysis, provided that the Phase-I inspection is performed immediately and no damage is found during this inspection.
- If a discrepancy caused by the hard landing issue is found during the Phase-I inspection, the Phase-II inspection must be done immediately.
- During the fly-by period, the Phase-II inspection must be done.

(g) If no discrepancy is found during the Phase-II inspections for the aircraft structure, it is cleared for serviced. But, if you find any discrepancy, proceed as follows:

1 Do the aircraft structure phase-III inspections and send Embraer these data;

a Flight data source that contain the hard landing incident;

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b Damage report with information about the damage found during the inspections.

2 Wait for Embraer disposition about the damage found and repair instructions before returning the aircraft to service.

3 Repair the aircraft structure if applicable.

4 After that, the aircraft structure is cleared for service.

(h) For the engines, if no serviceability threshold was exceeded according to GE manual, they are cleared for service. But, if you find any discrepancy, proceed as follows:

1 Check for fly-by instructions on engine manual and send findings to Engine Manufacturer.

2 If there is a fly-by granted engines are serviceable for the fly-by period.

3 Wait for engine manufacturer disposition about the damage found and repair instructions before returning the engines to service.

4 After that, the engines is cleared for service.

G. Hard-Landing Detection based on pitch rate monitor

SUBTASK 210-039-A

(1) Hard landing detection procedure based on pitch rate monitor is applicable when the aircraft hits hard the ground over the nose landing gear.

(2) The excess of the parameter threshold will classify the landing as hard landing.

NOTE:
- There are no thresholds for the engines based on the pitch rate monitor. However if for some reason the aircraft structure phase-III inspection is required, engines inspection phase-II are mandatory.
- On the hard landing analysis if the landing gear that may have exceeded the envelope is not known or there are doubts about the event, make sure to do the analyses based on all hard landing monitors.
- When the crew members report a possible hard landing and the flight data will be analyzed, make sure to do the analyses based on all hard landing monitors.
- In situations when the aircraft touches the ground with the NLG first, none of hard landing detection procedures using flight data are applicable and phase-II inspections for the aircraft structure are required.

(3) *EFFECTIVITY: ON ACFT WITH HONEYWELL DVDR OR PRE-MOD SB 170-31-0050/10*

To do the landing detection based on the pitch rate monitor, it is necessary to download the flight data from DVDR 1 or DVDR 2 or QAR. Refer to AMM TASK 31-31-00-970-801-A/200 or AMM TASK 31-31-00-970-802-A/200 or , as necessary.

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(4) *EFFECTIVITY: ON ACFT WITH UNIVERSAL DVDR OR POST-MOD SB 170-31-0050/10*

To do the landing detection based on the pitch rate monitor, it is necessary to download the flight data from DVDR 1 or DVDR 2 or QAR. Refer to AMM TASK 31-31-00-970-804-A/200 or , as necessary.

(5) The steps described below for hard landing analysis based on the pitch rate monitor are summarized on the flowchart from Figure 604.

(6) In the flight data, mark the initial and the final data samples to specify the range for pitch data for analysis (Figure 606):

(a) The range starts at the first pitch angle (Pitch) sample with value less than 5.0 degrees after the maximum vertical CG acceleration recorded in the range specified as before vertical acceleration monitor.

(b) The range stops at the first pitch angle (Pitch) sample with value lower than or equal to 1.0 degree.

(7) Calculate the pitch rates as shown in Figure 609.

(8) With the maximum pitch rate calculated determine the actions to be taken based on the instructions below:

(a) If the maximum pitch rate recorded is greater than or equal to the threshold of -6.00 degrees per second, the aircraft is cleared for service based on the pitch rate monitor.

(b) If the maximum pitch rate recorded is less than (exceeds) the threshold of -6.00 degrees per second, do the Phase-II inspection indicated for the aircraft structure when the pitch rate monitor is exceeded.

NOTE:
- In the example shown in Figure 609, the pitch rate value of -4.4 degrees per second is more than than the threshold value.
- Make sure that the Phase-I inspection is performed before starting the Phase-II.
- If it is not possible to do Phase-II inspection immediately, refer to Figure 617 for fly-by instructions. The fly-by is granted from the moment of the hard landing detection through the flight data analysis, provided that the Phase-I inspection is performed immediately and no damage is found during this inspection.
- If a discrepancy caused by the hard landing issue is found during the Phase-I inspection, the Phase-II inspection must be done immediately.
- During the fly-by period, the Phase-II inspection must be done.

(9) If no discrepancy is found during the phase-II inspections for the aircraft structure, the aircraft is cleared for service. But, If you find any discrepancy, proceed as follows:

(a) Do the Phase-III inspection indicated for the aircraft structure when the pitch rate monitor is exceeded and send Embraer these data:

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1 Flight Data Source that contain the hard landing incident;

2 Damage report with information about the damage found during the inspections.

(b) Wait for Embraer disposition about the damage found and repair instructions before returning the aircraft to service.

(c) Repair the aircraft if applicable.

(d) After that, the aircraft structure is cleared for service.

(10) In case aircraft structure Phase-III inspection was necessary for some reason, perform compulsorily the Phase-II inspection indicated for the engine according to GE Engine Manual overlimit condition Inspection LMM TASK 72-00-00-220-801 (Subtask 72-00-00-280-002).

(a) If no serviceability threshold was exceeded according to GE Manual, engines are cleared for service. But, If you find any discrepancy, proceed as follows:

1 Check for fly-by instructions on engine manual and send findings to Engine Manufacturer.

2 If there is a fly-by granted engines are serviceable for the fly-by period.

3 Wait for engine manufacturer disposition about the damage found and repair instructions before returning the engine to service.

4 After that, the engine is cleared for service.

H. Job Set-Up

SUBTASK 940-020-A

NOTE: When Phase-I, Phase-II, or Phase-III inspections are requested based on a given monitor do only the inspections required for that monitor. However, if no monitor is associated to the inspection request do Phase I, Phase II or Phase III inspections requested for all monitors.

(1) Do the procedure to make the aircraft safe for maintenance (AMM TASK 20-00-00-910-801-A/200).

SUBTASK 940-021-A

(2) Set all flight control surfaces to the neutral position.

I. Phase-I Inspection - For vertical acceleration monitor and/or roll rate monitor exceedance

SUBTASK 210-040-A

NOTE: Do the following subtask inspections only if they have been called by one of the hard landing detection procedures or off-temperature envelope procedure.

(1) Visually examine the MLG items but do not disassemble them.

NOTE: The MLG disassembly is necessary, if you think that there is damage related to a hard landing, after an external inspection.

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(a) Examine the MLG wheels for distortion, bending, dents, and fracture.

(b) Examine the MLG tires for general condition and deflation.

(c) Examine the brake assembly for damage and hoses for signs of fluid leakage.

(d) Examine the MLG wheelwell for signs of fluid leakage.

(e) Examine the MLG locking stay for distortion, bending, and fracture.

(f) Examine the MLG torque link for distortion, bending, and fracture.

(g) Examine the upper and lower MLG side stays for distortion, bending, and fracture.

(h) Examine the MLG strut for leakage around the sliding tube diameter.

(2) Fuselage

(a) Examine the external surface of the fuselage for fluid leakage.

(b) Externally examine the areas near the wing-stub for signs of damage.

(c) Open and close the aft passenger door and make sure that it operates correctly (AMM SDS 52-12-00/1).

(d) Open and close the aft service door and make sure that it operates correctly (AMM SDS 52-42-00/1).

(e) Open and close the forward and aft cargo doors and make sure that they operate correctly (AMM SDS 52-31-00/1 and AMM SDS 52-32-00/1).

(f) Examine the tail cone and silencer.

(g) Examine the wing-to-fuselage fairing junction with the fuselage and look for missing or damaged screw, missing sealant, and paint peel-off.

(3) Wing

(a) Examine the MLG doors.

(b) Examine the external surface of the wing for fuel or other fluid leakage.

NOTE: Carefully examine the fuel-tank access panel contours for fuel leakage.

(c) Examine the pylon fairings for signs of damage.

(4) Flight Controls

(a) With the flaps fully retracted, examine the flap surfaces for misalignment.

(b) With the flaps fully deflected, examine the flap rod mechanisms for bent components.

(c) Examine all flight controls to make sure that they move freely.

NOTE: If necessary use a flashlight to inspect inside the flap rod fairing.

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(d) Do an inspection of the flap rod mechanism (AMM TASK 57-51-00-200-801-A/600).

NOTE: If flap rods are bent but no other related parts are damaged, only flap rod replacement is necessary. In this case, if you don't have any other discrepancy during phase I inspection, phase II inspection is not necessary.

(5) Engine

(a) Examine the inlet-cowl internal-skin for signs of chafing against the fan blades.

J. Phase-II Inspection - For vertical acceleration monitor and/or roll rate monitor exceedance

SUBTASK 210-041-A

NOTE: Do the following subtask inspections only if they have been called by one of the hard landing detection procedures or off-temperature envelope procedure.

(1) Main Landing Gear

(a) Do an inspection on the lower skin of the rear fuselage around the jacking point before you jack the aircraft. If there are signs of buckling, do not put the aircraft on jacks and contact Embraer for more instructions.

(b) Put the aircraft on jacks (AMM TASK 07-10-01-500-801-A/200).

(c) Do a check of the gaps of the proximity sensors. The sensor surfaces must be parallel.

1 For downlock sensor, refer to AMM TASK 32-61-07-820-801-A/200.

2 For weight-on-wheels sensor, refer to AMM TASK 32-62-03-820-801-A/200.

(d) Remove the MLG side stay lower bolt (AMM TASK 32-11-05-000-801-A/400) and examine for distortion, bending, and fracture.

(e) Examine side stay lower bolt lug for distortion, bending, and fracture. Refer to Figure 611, Sheet 1.

(f) Install the MLG side stay lower sleeve (AMM TASK 32-11-05-400-801-A/400).

(g) Examine the trunnion fitting. Refer to Figure 611, Sheet 2.

(h) Connect the external AC power supply to the aircraft (AMM TASK 24-42-02-860-801-A/200).

(i) Retract the landing gear and visually check during the retraction path for interferences, gaps, misalignments, malfunctions and unusual noise in the landing gear, landing gear bay and landing gear door mechanisms. Extend landing gear and check for unusual noise, up lock and down lock mechanisms malfunction. Check for any abnormal behavior.

NOTE: An irregular, non-smooth operation of the landing gear could show deformation of the landing gear. You must then examine it for correct alignment.

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(j) Repeat the retraction procedure and extend the landing gear using the free-fall system. Check for any abnormal behavior.

NOTE: An irregular, non-smooth operation of the landing gear could show deformation of the landing gear. You must then examine it for correct alignment.

(k) Do an inspection of the main landing gear, doors and door linkages (AMM TASK 32-10-00-200-801-A/600).

NOTE: This inspection is necessary only if landing gears were operated by free-fall method, without pressurization of the hydraulic system.

(l) Disconnect the external AC power supply from the aircraft (AMM TASK 24-42-02-860-802-A/200).

(m) Remove the brake assembly (AMM TASK 32-49-11-000-801-A/400).

(n) Examine the wheel axle to know if it is bent.

(o) Install the brake assembly (AMM TASK 32-49-11-400-801-A/400).

(p) If tire bursts during the event, disassemble the torque links and check pins and lugs for distortion, bending and fracture. Examine the MLG shimmy damper (AMM TASK 32-11-03-200-802-A/600).

(q) Lower the aircraft from the jacks (AMM TASK 07-10-01-500-802-A/200).

(2) Fuselage

(a) Remove the forward panel of the wing-to-fuselage fairing (AMM TASK 53-04-20-000-801-A/400).

NOTE: As alternative step, you can remove only access panels 191AR, 191EL, 191BL, and 191CL and a boroscope inspection is done.

(b) Remove access panels 192BR, 192CR, 192AL, and 192BL (AMM TASK 06-41-00-800-801-A/100).

(c) Remove the side panel of the wing-to-fuselage fairing (AMM TASK 53-04-40-000-801-A/400).

(d) Remove the aft panel of the wing-to-fuselage fairing (AMM TASK 53-04-50-000-801-A/400).

NOTE: As alternative step, you can remove only access panels 195AL, 195BL, 195DL, 195AR, and 195CR, and a boroscope inspection is done.

(e) Externally examine the upper and lower fuselage skin panels for loose or sheared rivets, structural damage and signs of fuel or other fluid leakage. If you find wrinkles on the skin panels, do an internal visual inspection of the fuselage.

(f) Carefully examine all equipment inside the wing-to-fuselage fairing region and its attachment to the fuselage.

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(g) In the forward avionics compartment, examine all equipment, their racks, shelves and supports for integrity.

(h) In the middle avionics compartment, examine all equipment, their racks, shelves and supports for integrity.

(i) In the rear fuselage, examine all equipment, their racks, shelves and supports for integrity.

(j) Carefully examine the internal side of the fuselage in the attachment of the galley to the frame. Refer to Figure 610, Sheet 1, Figure 610, Sheet 2 and/or Figure 610, Sheet 3.

NOTE: It is not necessary to remove the galley to get access to the inspection area.

(k) In the rear fuselage, examine the potable water tank and its attachment to the fuselage structure.

(l) In the rear fuselage, examine the vacuum waste-system tank and its attachment to the fuselage structure.

(m) Install the aft panel of the wing-to-fuselage fairing (AMM TASK 53-04-50-400-801-A/400).

(n) Install the side panel of the wing-to-fuselage fairing (AMM TASK 53-04-40-400-801-A/400).

(o) Install access panels 192BR, 192CR, 192AL, and 192BL (AMM TASK 06-41-00-800-801-A/100).

(p) Install the forward panel of the wing-to-fuselage fairing (AMM TASK 53-04-20-400-801-A/400).

(3) Wing

(a) Examine wing spars II and III. Start from the MLG wheelwell. Remove the access panels and, with the inboard flap down, also examine the lower skin adjacent to them.

(b) Examine the upper and lower wing skin panels from the wing root to the wing tip.

(c) Externally examine the wing, nacelles, and engine struts for loose or sheared rivets, structural damage, and signs of fuel and other fluid leakage.

(d) Examine the wing-to-fuselage junction and the MLG wheelwell for damage. Look for flaked paint and pulled-out or missing fasteners.

(4) Flight Controls

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(a) Do an inspection of the flap rod mechanism (AMM TASK 57-51-00-200-801-A/600).

NOTE: If flap rods are bent but no other related parts are damaged, only flap rod replacement is necessary. In this case, if you don't have any other discrepancy during phase II inspection, phase III inspection is not necessary.

(b) With the flaps and slats fully deflected, examine the flap and slat mechanisms for bent components.

(c) With the flaps and slats fully retracted, examine the flap and slat surfaces for misalignment.

(d) In the MLG compartment, do a check to see if the aileron cable and the pulleys are correctly installed.

(5) Engine

(a) NOTE: This engine inspection is only needed if the engine threshold is exceeded (Figure 607) or if the landing was done below -45°C.

Do the GE Engine Manual overlimit condition inspection LMM TASK 72-00-00-220-801 (Subtask 72-00-00-280-002).

(6) APU

(a) Do a visual inspection of the APU external side.

(b) Start the APU, then operate it for 15 minutes minimum.

(c) Shut down the APU normally, then examine the magnetic drain plug for debris (AMM TASK 49-91-05-200-801-A/600).

K. Phase-III Inspection - For vertical acceleration monitor and/or roll rate monitor exceedance

SUBTASK 210-042-A

NOTE: Do the following subtask inspections only if they have been called by one of the hard landing detection procedures or off-temperature envelope procedure.

(1) Make sure you have completely done Phase-II inspections.

(2) Inspect the rear fuselage lower skin around the jacking point before you jack the aircraft. If there are signs of buckling, do not put the aircraft on jacks and contact Embraer for further instructions.

(3) Put the aircraft on jacks (AMM TASK 07-10-01-500-801-A/200).

(4) Remove the MLG locking stay pins (AMM TASK 32-11-07-000-801-A/400) and examine for distortion, bending, and fracture.

(5) Remove the MLG torque link pins (AMM TASK 32-11-11-000-801-A/400) and examine for distortion, bending, and fracture.

(6) Remove the MLG pintle pins (AMM TASK 32-11-01-000-801-A/400) and examine for distortion, bending, and fracture.

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(7) Do GVI inspection of the CF II internal side in the area below the floor cross beams and columns. Refer to Figure 618.

(8) Do the aircraft alignment check (AMM TASK 51-50-00-800-801-A/600).

(9) In case it has not been done yet, do the GE Engine Manual overlimit condition Inspection LMM TASK 72-00-00-220-801 (Subtask 72-00-00-280-002).

L. Phase-I Inspection - For pitch rate monitor exceedance

SUBTASK 210-043-A

NOTE: Do the following subtask inspections only if they have been called by one of the hard landing detection procedures or off-temperature envelope procedure.

(1) Visually examine the items that follow. It is not necessary to disassemble them:

NOTE: The NLG disassembly is only necessary if you think that there is damage related to the hard landing after an external examination.

(2) Nose Landing Gear

(a) Examine the NLG wheels for bend, fracture, dents, and wear.

(b) Examine the NLG tires for general condition and deflation.

(c) Examine the NLG strut for leakage around the sliding tube diameter.

(d) Examine the steering cylinder for signs of fluid leaks.

(e) Examine the NLG wheelwell for signs of fluid leaks.

(f) Examine the NLG locking stay for distortion, bending, and fracture.

(g) Examine the NLG torque links for distortion, bending, and fracture.

(h) Examine the NLG upper and lower drag braces for distortion, bending, and fracture.

(i) Examine the hydraulic actuators for bend, fracture of rod ends, and fluid leakage.

(3) Fuselage

(a) Examine the external surface of the fuselage for fluid leakage.

(b) Examine the NLG bay for signs of fluid leakage or distortion.

(c) Externally examine the lower fuselage skin panels.

NOTE: Carefully examine the rivets for integrity at the contour of the NLG bay.

(d) Examine the NLG doors and the rods for distortion.

(e) Examine the NLG attachment fittings.

(f) Open and close the forward passenger door and make sure that it operates correctly (AMM SDS 52-11-00/1).

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(g) Open and close the forward service door and make sure that it operates correctly (AMM SDS 52-41-00/1).

M. Phase-II Inspection - For pitch rate monitor exceedance

SUBTASK 210-044-A

NOTE: Do the following subtask inspections only if they have been called by one of the hard landing detection procedures or off-temperature envelope procedure.

(1) Nose Landing Gear

(a) Examine the lower skin of the rear fuselage around the jacking point before you lift the aircraft. If there are signs of buckling, do not put the aircraft on jacks and contact Embraer for more instructions.

(b) Put the aircraft on jacks (AMM TASK 07-10-01-500-801-A/200).

(c) Do a check of the gaps of the proximity sensors. The sensor surfaces must be parallel.

1 For downlock sensor, refer to AMM TASK 32-61-03-820-801-A/200.

2 For weight-on-wheels sensor, refer to AMM TASK 32-62-01-820-801-A/200.

(d) Examine the NLG attachment fittings.

(e) Retract the landing gear and visually check during the retraction path for interferences, gaps, misalignments, malfunctions and unusual noise in the landing gear, landing gear bay and landing gear door mechanisms. Extend landing gear and check for unusual noise, up lock and down lock mechanisms malfunction. Check for any abnormal behavior.

NOTE: An irregular, non-smooth operation of the landing gear could show deformation of the landing gear. You must then examine it for correct alignment.

(f) Repeat the retraction procedure and extend the landing gear using the free-fall system. Check for any abnormal behavior.

NOTE: An irregular, non-smooth operation of the landing gear could show deformation of the landing gear. You must then examine it for correct alignment.

(g) Do an inspection of the nose landing gear, doors and door linkages (AMM TASK 32-20-00-200-801-A/600).

NOTE: This inspection is necessary only if landing gears were operated by free-fall method, without pressurization of the hydraulic system.

(h) Examine the drag-and-locking-strut link assemblies and drag-and-locking-strut support fitting.

(i) Examine the NLG wheelwell for buckled structure.

(j) Examine the NLG axle (AMM TASK 32-49-01-000-801-A/400) to see if it is cracked or bent.

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(2) Fuselage

(a) Open these access panels to get access to the internal side of the NLG wheelwell:

- 125AZ
- 126AZ

(b) Carefully examine the gray areas shown in Figure 613 for signs of damage or distortion.

(c) Examine the fuselage skin attachments above the landing gear beam for distortion. Look for flaked paint and pulled-out or missing fasteners.

(d) In the forward avionics compartment, examine all equipment, their racks, shelves, and supports for integrity.

(e) In the radome, do a check to see if the antennas are in position and examine the web structure for integrity.

(f) Do the RAT deployment operational test (AMM TASK 24-23-00-710-801-A/500).

N. Phase-III Inspection - For pitch rate monitor exceedance

SUBTASK 210-045-A

NOTE: Do the following subtask inspections only if they have been called by one of the hard landing detection procedures or off-temperature envelope procedure.

(1) Make sure you have completely done Phase-II inspections.

(2) Inspect the rear fuselage lower skin around the jacking point before you put the aircraft on jacks. If there are signs of buckling, do not put the aircraft on jacks and contact Embraer for further instructions.

(3) Put the aircraft on jacks (AMM TASK 07-10-01-500-801-A/200).

(4) Remove the NLG locking stay pins (AMM TASK 32-21-05-000-801-A/400) and examine for distortion, bending, and fracture.

(5) Remove the NLG upper and lower drag-brace pins (AMM TASK 32-21-03-000-801-A/400) and examine for distortion, bending, and fracture.

(6) Do the aircraft alignment check (AMM TASK 51-50-00-800-801-A/600).

(7) Do the GE Engine Manual overlimit condition Inspection LMM TASK 72-00-00-220-801 (Subtask 72-00-00-280-002).

O. Job Close-Up

SUBTASK 940-022-A

(1) Put the aircraft back to its initial condition (AMM TASK 20-00-00-910-801-A/200).

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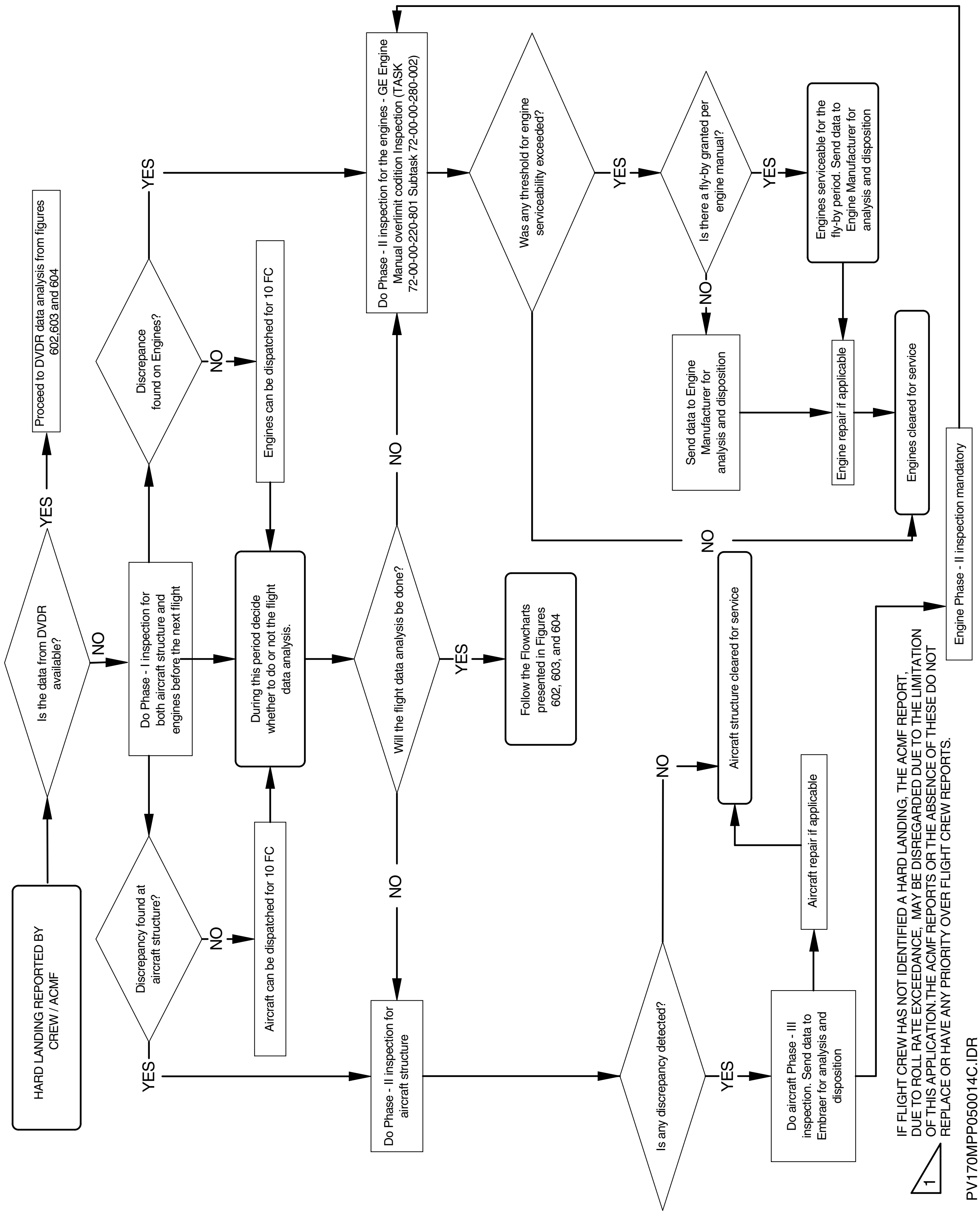
EFFECTIVITY: ALL

Hard Landing Flowchart - Crew Report

Figure 601



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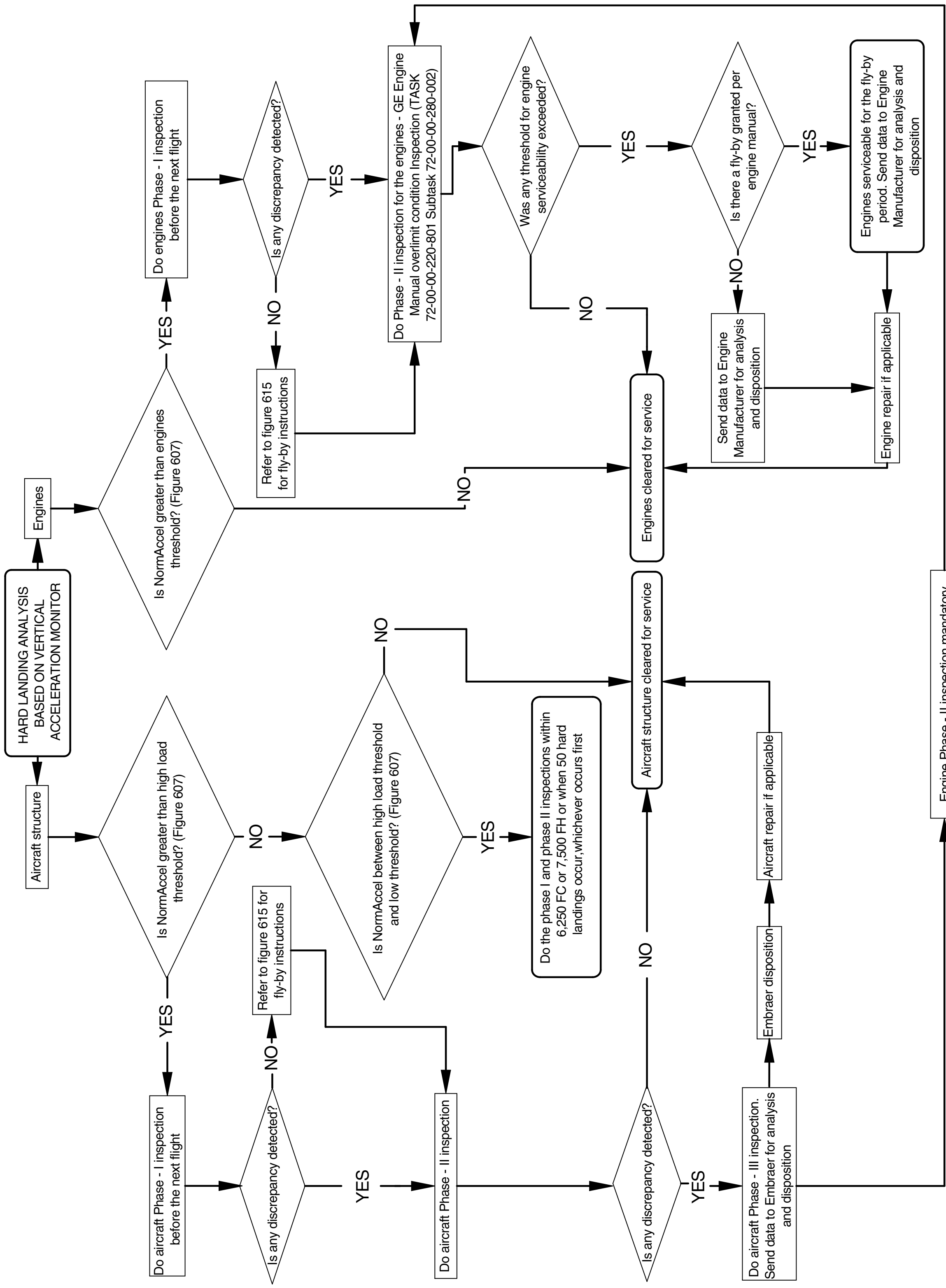
EFFECTIVITY: ALL

Hard Landing Flowchart - Flight Data Analysis - Hard landing analysis based on vertical acceleration monitor
Figure 602



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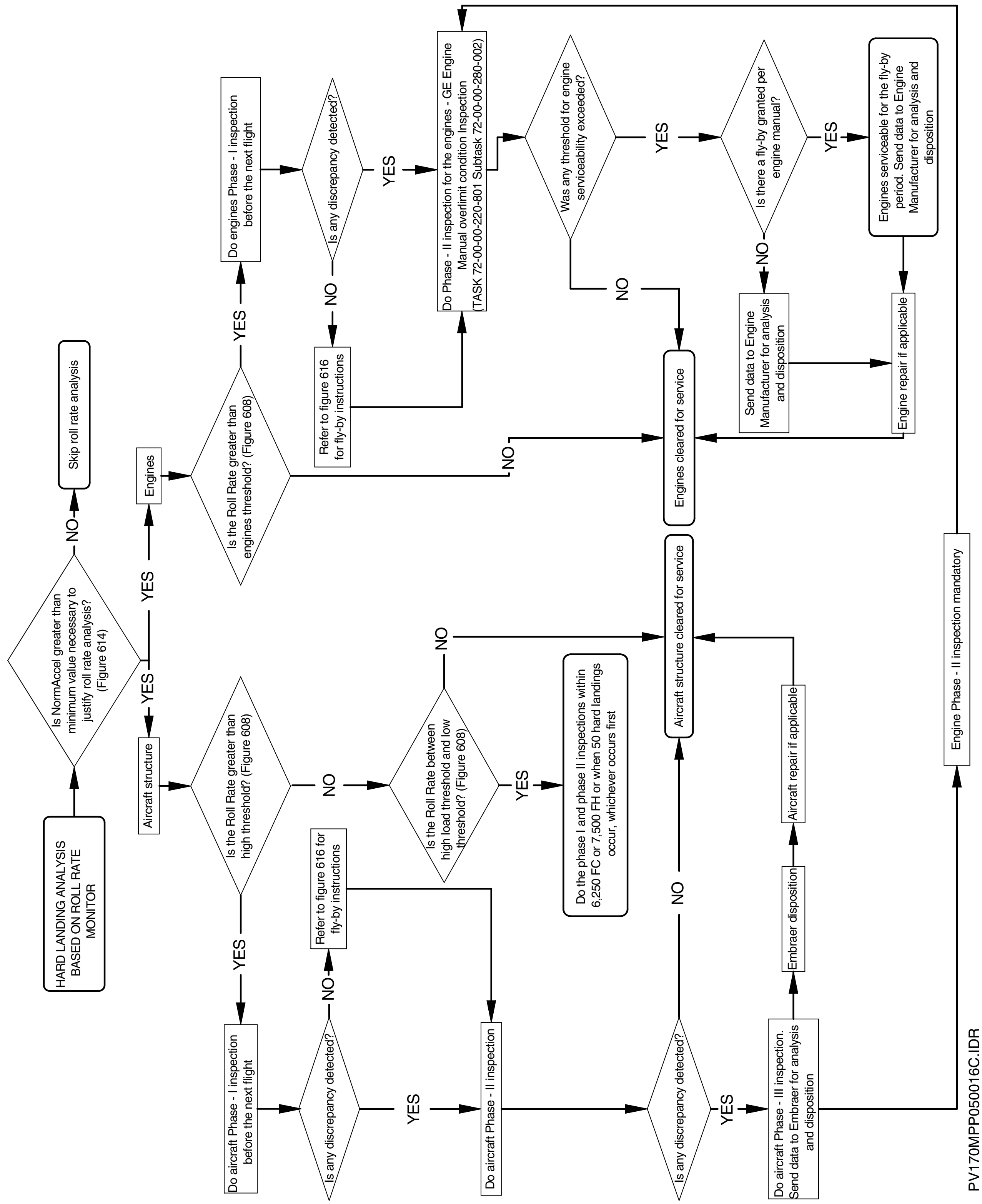
EFFECTIVITY: ALL

Hard Landing Flowchart - Flight Data Analysis - Hard landing analysis based on roll rate monitor

Figure 603



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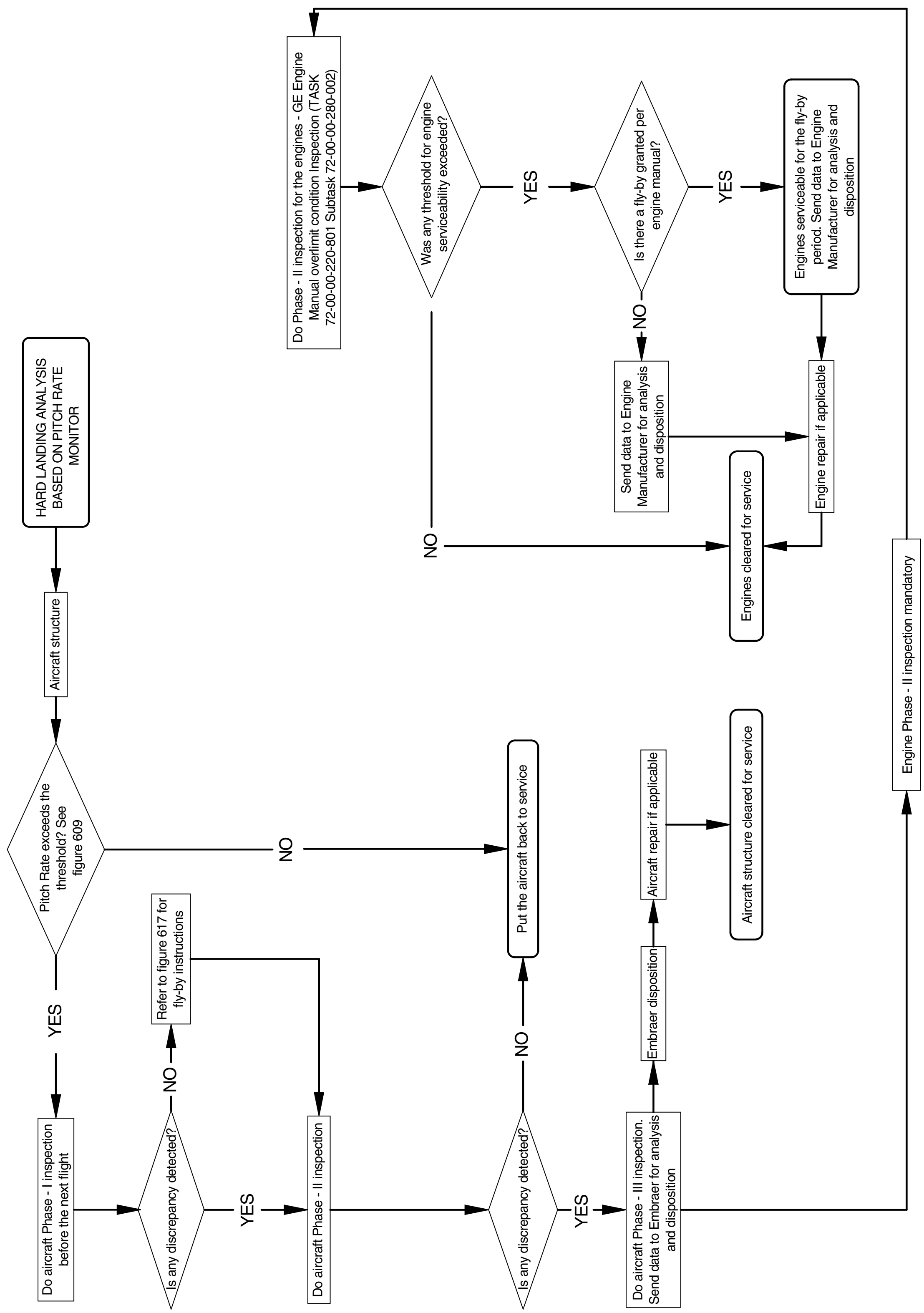
EFFECTIVITY: ALL

Hard Landing Flowchart - Flight Data Analysis - Hard landing analysis based on pitch rate monitor
Figure 604



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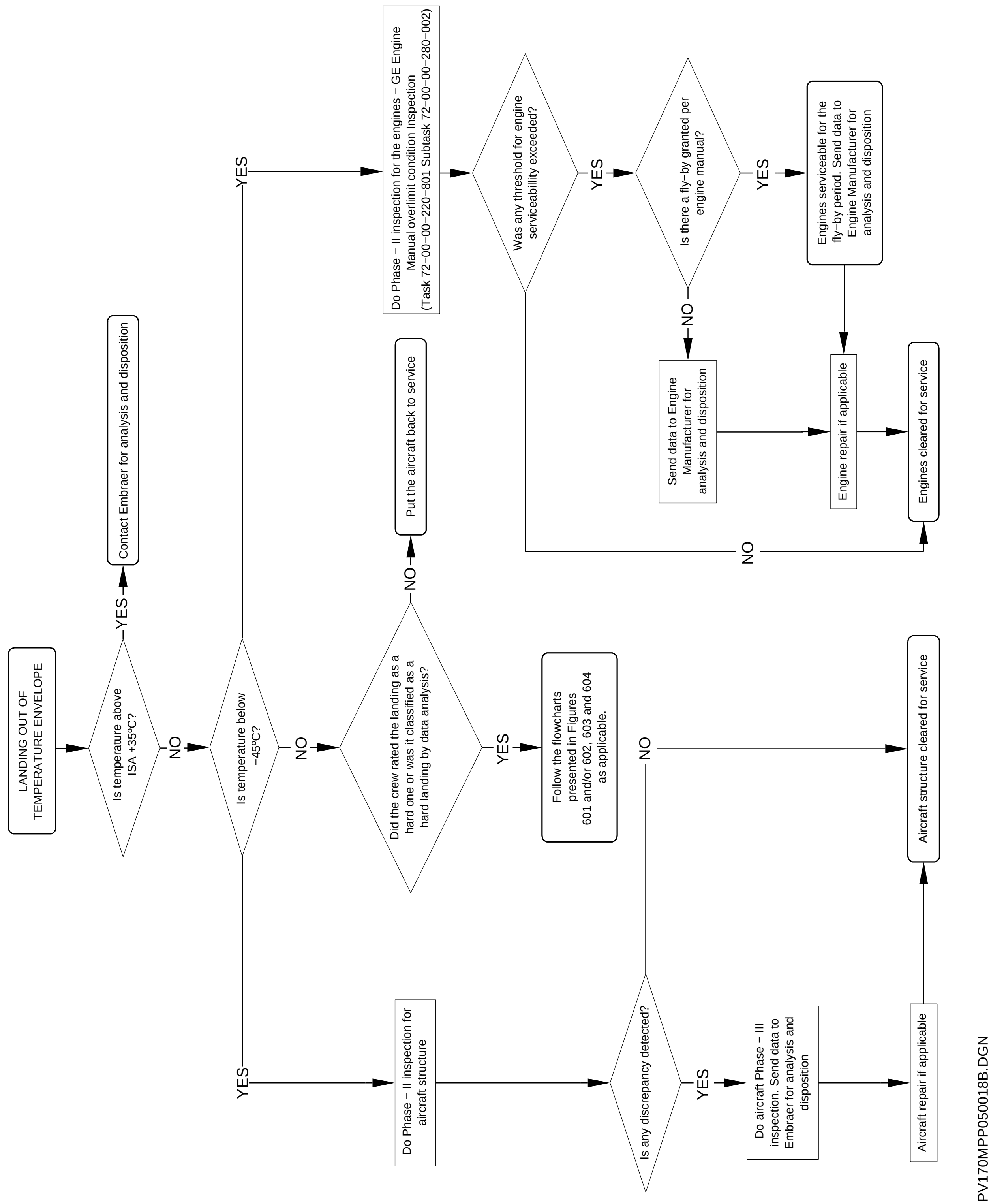
EFFECTIVITY: ALL

Hard Landing Flowchart - Out of Temperature Envelope

Figure 605



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EFFECTIVITY: ALL

Downloaded DVDR Data

Figure 606

Sample	GMT	GrossWeight (lb)	Air/Gnd	Roll (deg)	Pitch (deg)	NormAccel (g)
92172					7.9	1.063
92172						1.063
92173	09:51:32	70850	AIR	−1.2	7.7	1.051
92173						1.047
92173					7.4	1.066
92173						1.051
92173				−0.9	7	0.996
92173						0.984
92173					6.7	0.973
92173						0.969
92174	09:51:33		AIR	−1.4	6.5	0.996
92174						0.996
92174					6.3	0.98
92174						0.969
92174				−2.1	6.3	0.953
92174						0.957
92174					6.5	0.977
92174						0.977
92175	09:51:34		AIR	−2.6	6.7	0.977
92175						0.992
92175					6.9	1.016
92175						1.004
92175				−3.2	6.9	1.004
92175						1.012
92175					7	0.992
92175						1.184
92176	09:51:35		AIR	−0.5	6.9	1.277
92176						1.137
92176					6.5	1.25
92176						1.242
92176				2.3	6	1.059
92176						0.91
92176					5.8	0.82
92176						0.852
92177	09:51:36	70850	GROUND	0.9	6.2	0.891
92177						0.973
92177					6.2	1.152
92177						1.355
92177				0.7	5.8	1.297
92177						1.105
92177					5.4	0.887
92177						0.883
92178	09:51:37		GROUND	0.7	4.9	0.93
92178						0.961
92178					4.4	1.023
92178						1.082
92178				0.5	3.5	1.121
92178						1.07
92178					2.6	0.914
92178						1
92179	09:51:38		GROUND	0.5	1.8	1.063
92179						1.07
92179					1.2	1.035
92179						1
92179				0.5	1.1	1.008
92179						1.047
92179					1.1	1.066
92179						0.992
92180	09:51:39		GROUND	0.7	1.1	0.98
92180						0.992
92180					0.9	1.027
92180						1.004
92180				0.7	0.7	0.996
92180						1.016
92180					0.5	1.02

A
B
C
D
E

LEGEND:

A − 4 SECONDS BEFORE THE FIRST "GROUND" RECORDED.
B − UP TO THE FIRST SAMPLE WITH PITCH LOWER THAN 1.0 DEGREE.
C − 2 SECONDS BEFORE THE MAXIMUM NORMAL ACCELERATION RECORDED.
D − 2 SECONDS AFTER THE MAXIMUM NORMAL ACCELERATION RECORDED.
E − FROM THE FIRST SAMPLE WITH PITCH LOWER THAN 5, UP TO THE FIRST SAMPLE WITH PITCH LOWER THAN 1.0 DEGREES.

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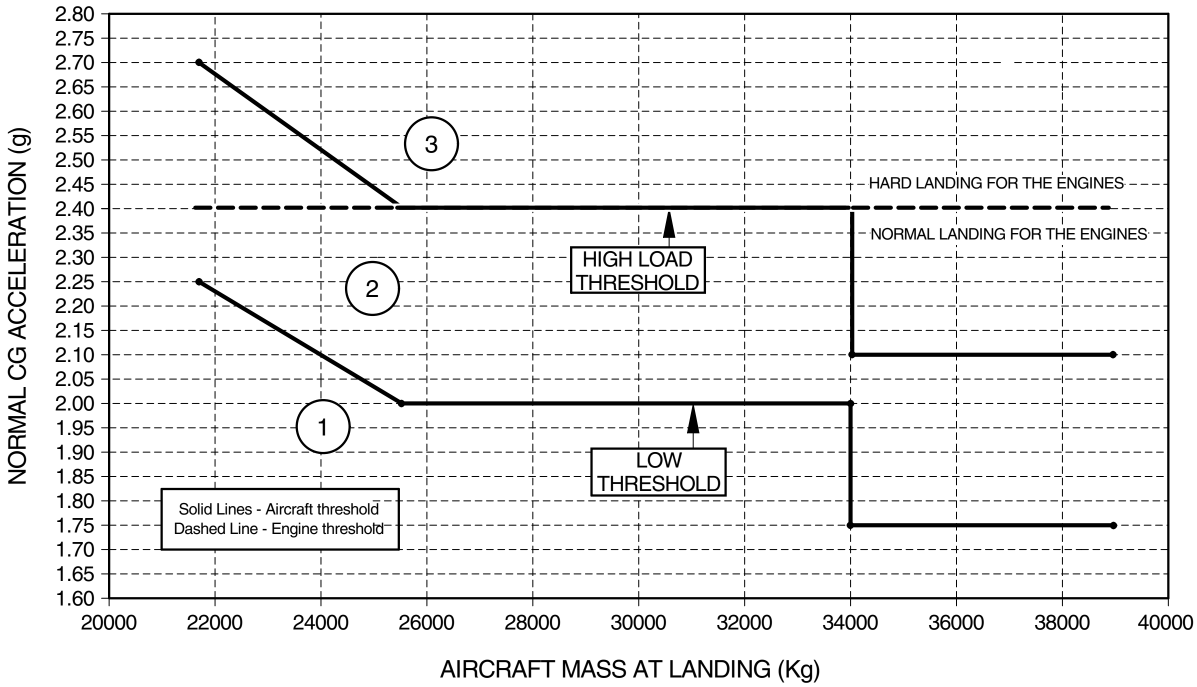
EFFECTIVITY: ALL
Hard Landing Thresholds
Figure 607



AIRCRAFT MASS (Kg/lb)	LOW THRESHOLD Nz (g)	20% HIGHER THRESHOLD Nz (g)
21800/48069	2.25	2.70
25500/56228	2.00	2.40
34000/74957	2.00	2.40
34000/74957	1.75	2.10
38790/85517	1.75	2.10

1 NORMAL LANDING.

2 DO THE PHASE - I AND II - INSPECTION AT THE NEXT SCHEDULED MAINTENANCE OPORTUNITY (NOT EXCEEDING 6,250 FC OR 7,500 FH) OR WHEN 50 HARD LANDING EVENTS OCCUR, WHICHEVER OCCURS FIRST.

3 DO THE PHASE - I - INSPECTION BEFORE NEXT FLIGHT.
DO THE PHASE - II - INSPECTION, REFER TO FIGURE 615 FOR PHASE - II - FLY-BY INSTRUCTIONS.

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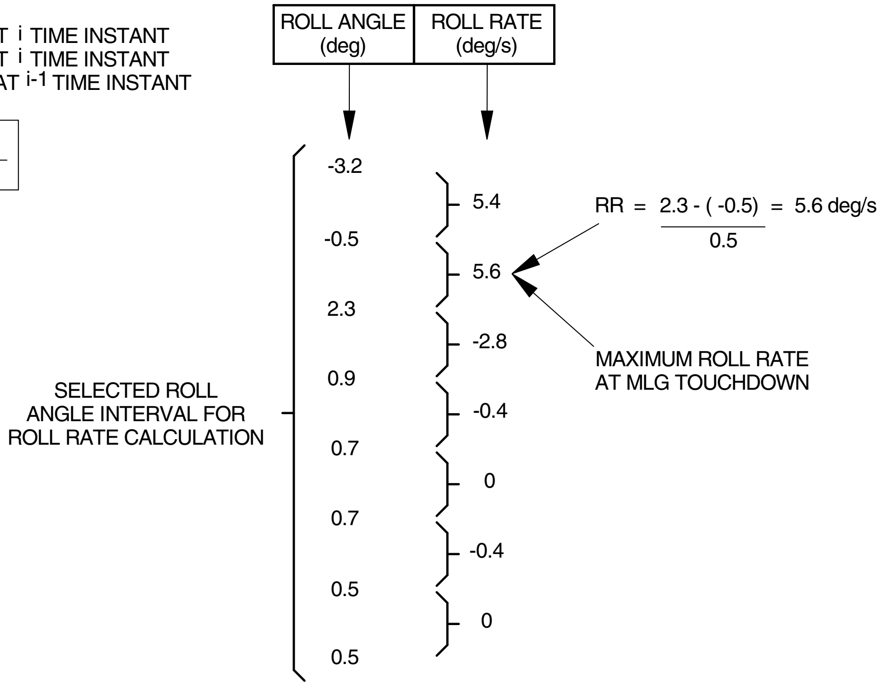
EFFECTIVITY: ALL

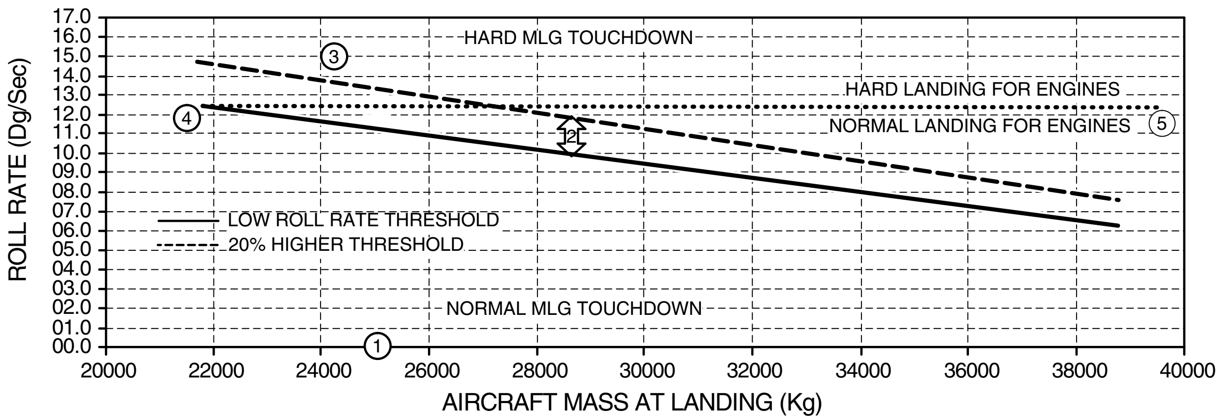
Roll Rate Calculation And Threshold
Figure 608

WHERE:

RR_i = ROLL RATE AT i TIME INSTANT
R_i = ROLL ANGLE AT i TIME INSTANT
R_{i-1} = ROLL ANGLE AT $i-1$ TIME INSTANT

$$RR_i = \frac{R_i - R_{i-1}}{0.5}$$





AIRCRAFT MASS (Kg)	LOW ROLL RATE THRESHOLD (Dg/Sec)	20% HIGHER ROLL RATE THRESHOLD (Dg/Sec)
38790	6,3	7,6
21800	12,4	14,9

NOTE 1: NORMAL LANDING

NOTE 2: DO THE PHASE I AND II INSPECTON AT A SCHEDULED MAINTENANCE OPPORTUNITY (NOT EXCEEDING 6,250 FC OR 7,500 FH) OR WHEN 50 HARD LANDING EVENTS OCCUR, WHICHEVER OCCURS FIRST

NOTE 3: DO THE PHASE - I - INSPECTON BEFORE NEXT FLIGHT

NOTE 4-5: ENGINE THRESHOLD 12.4 Dg/sec

NOTE 6: DO THE PHASE - II - INSPECTON, REFFER TO FIGURE 616 FOR PHASE - II - FLY-BY INSTRUCTIONS.

NOTE 7: USE ABSOLUTE VALUES TO DO THE CALCULATION

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EFFECTIVITY: ALL

Pitch Rate Calculation

Figure 609

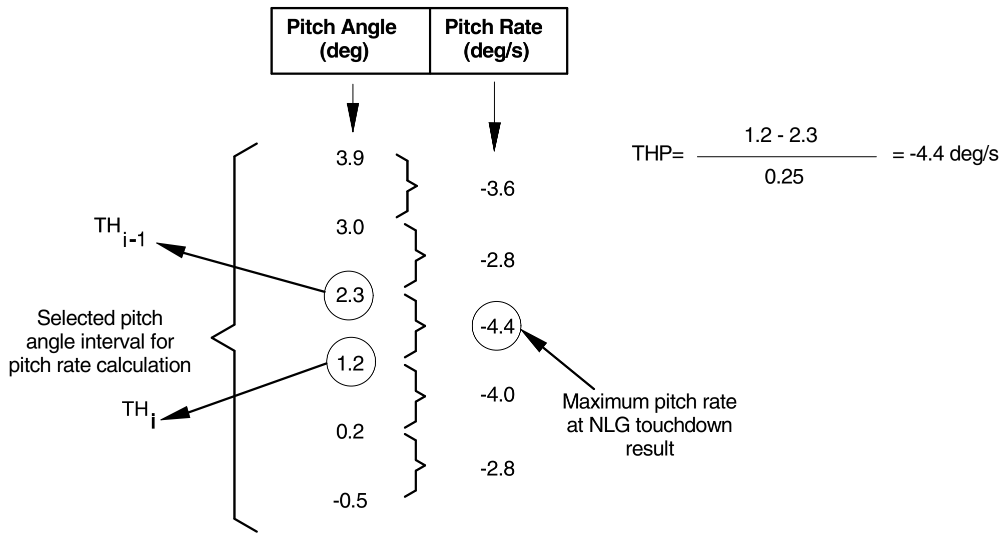
$$THP_i = \frac{TH_i - TH_{i-1}}{0.25}$$

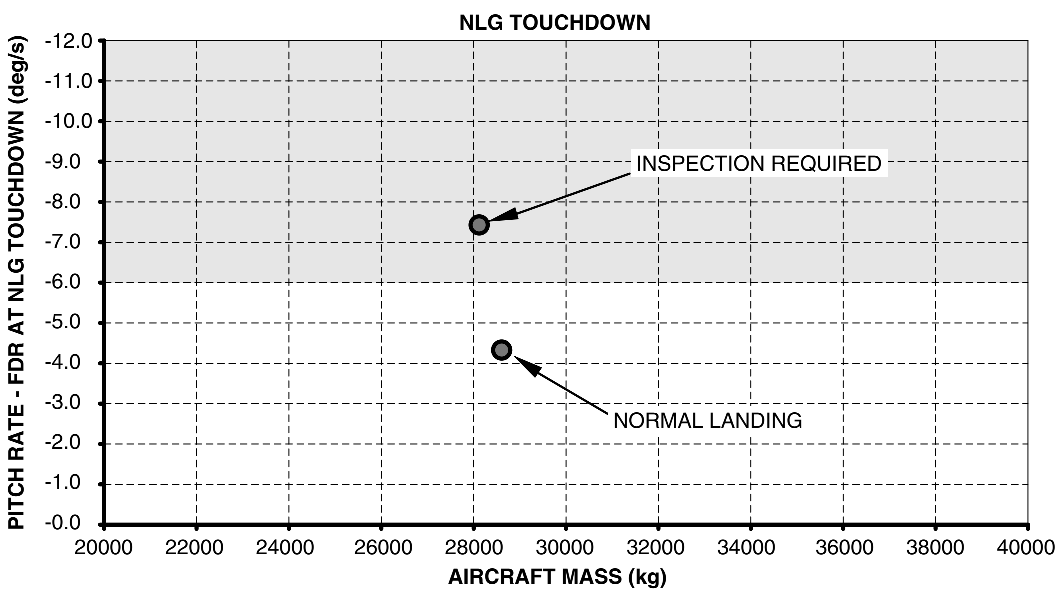
Where:

THP_i = pitch rate at i time instant;

TH_i = pitch angle at i time instant;

TH_{i-1} = pitch angle at i-1 time instant.





PICTH RATE THRESHOLD DETERMINATION

NOTE 1: THE PITCH RATE THRESHOLD IS -6.00 DEGREES PER SECOND.

NOTE 2: IF PITCH RATE IS AT OR ABOVE THE THRESHOLD, DO THE PHASE I AND PHASE II INSPECTION.

NOTE 3: IF PITCH RATE IS BELOW THE THRESHOLD, THE AIRCRAFT IS CLEARED FOR SERVICE.

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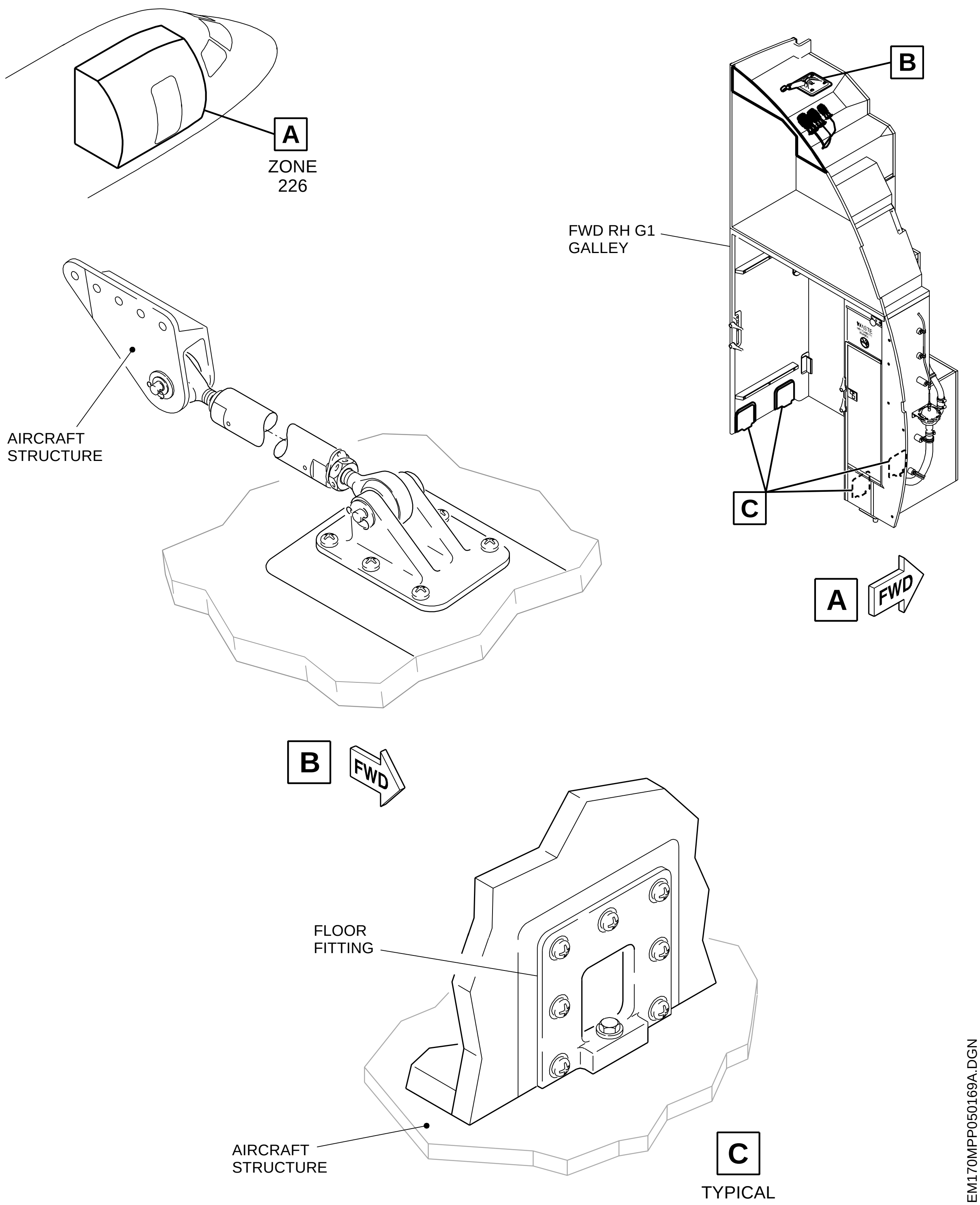
Galley Attachments - Inspection

Figure 610 - Sheet 1



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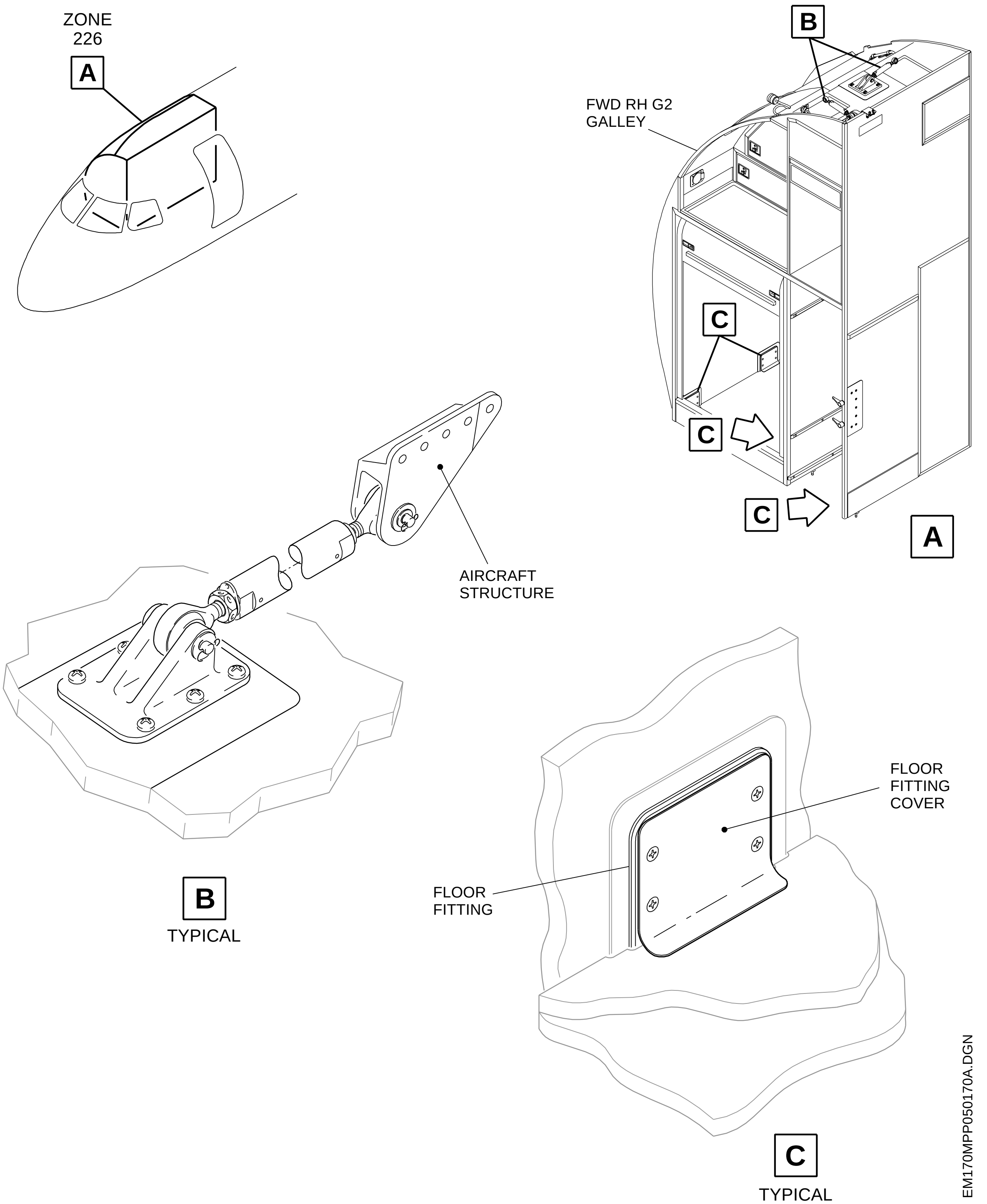
EFFECTIVITY: ALL
Galley Attachments - Inspection
Figure 610 - Sheet 2



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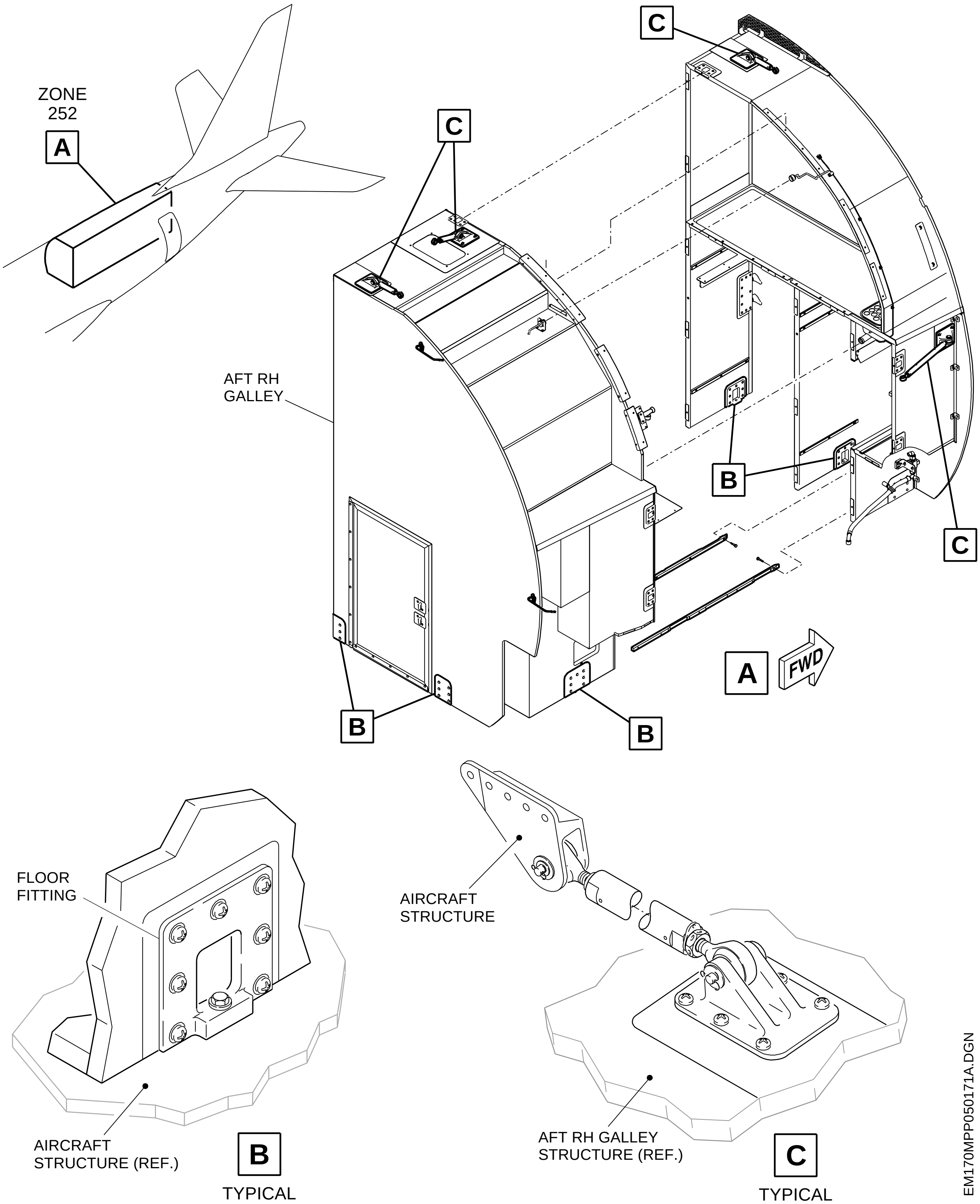
EFFECTIVITY: ALL
Galley Attachments - Inspection
Figure 610 - Sheet 3



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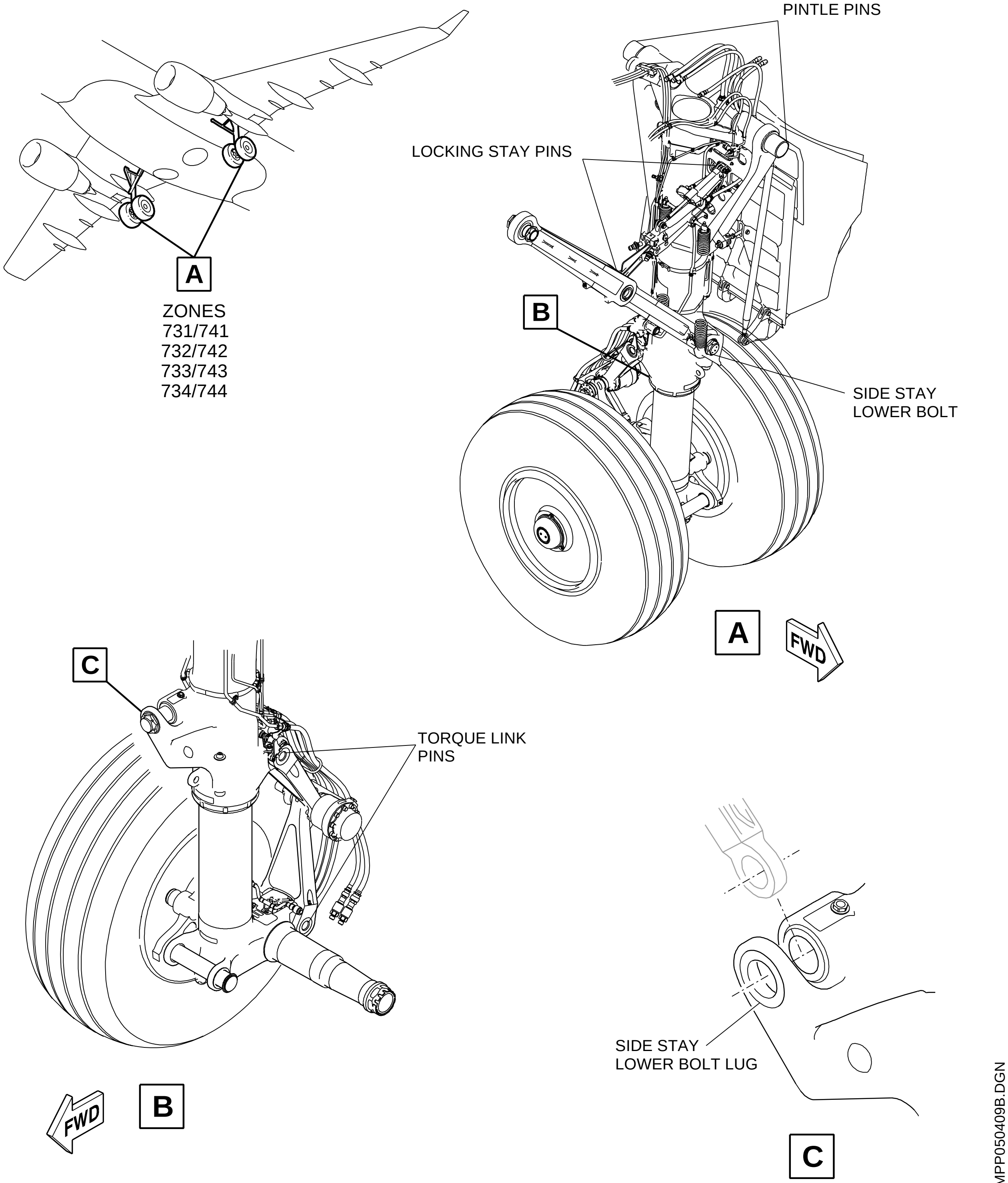
EFFECTIVITY: ALL

MLG Pins Identification and Structures

Figure 611 - Sheet 1



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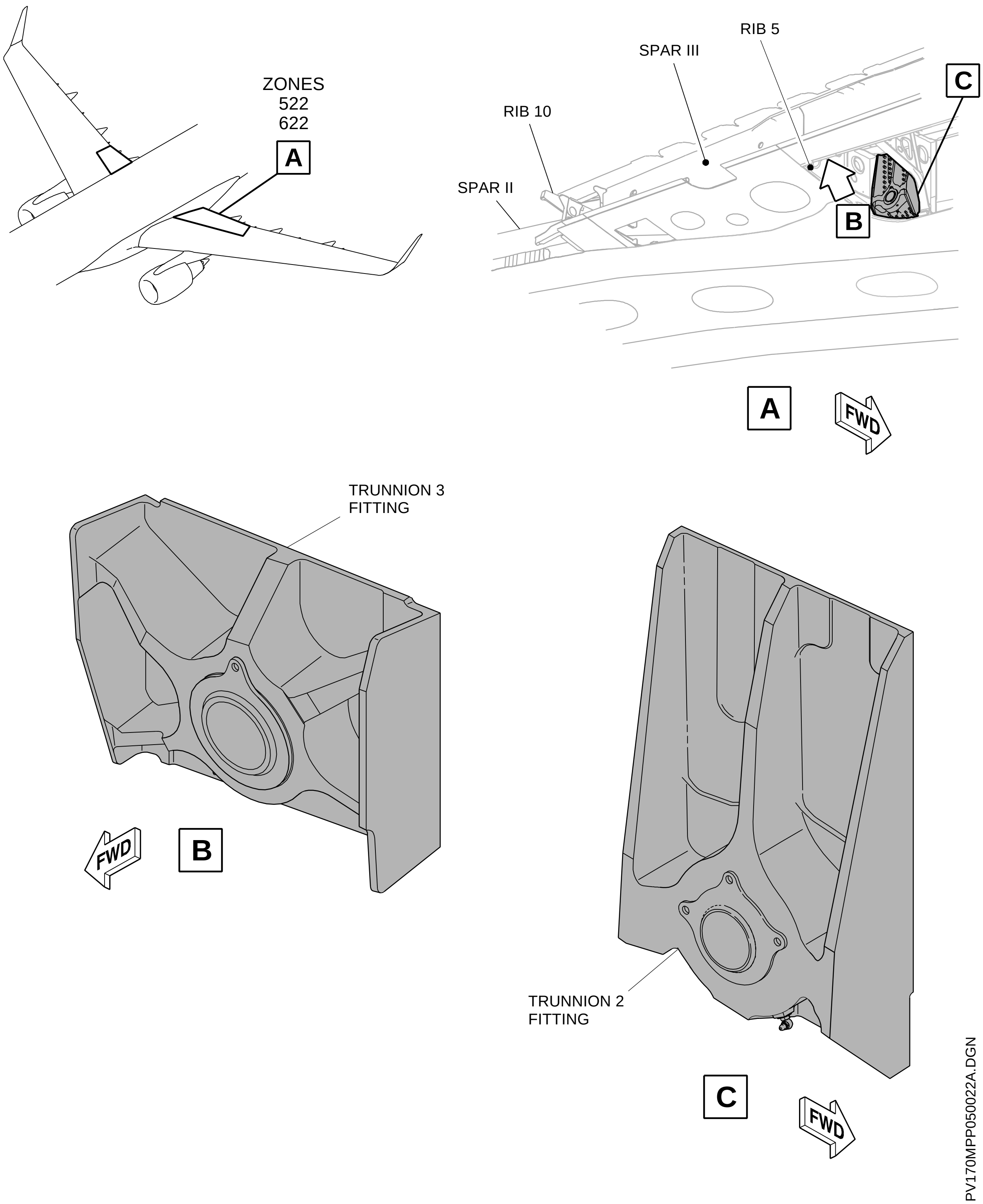
EFFECTIVITY: ALL

MLG Pins Identification and Structures

Figure 611 - Sheet 2



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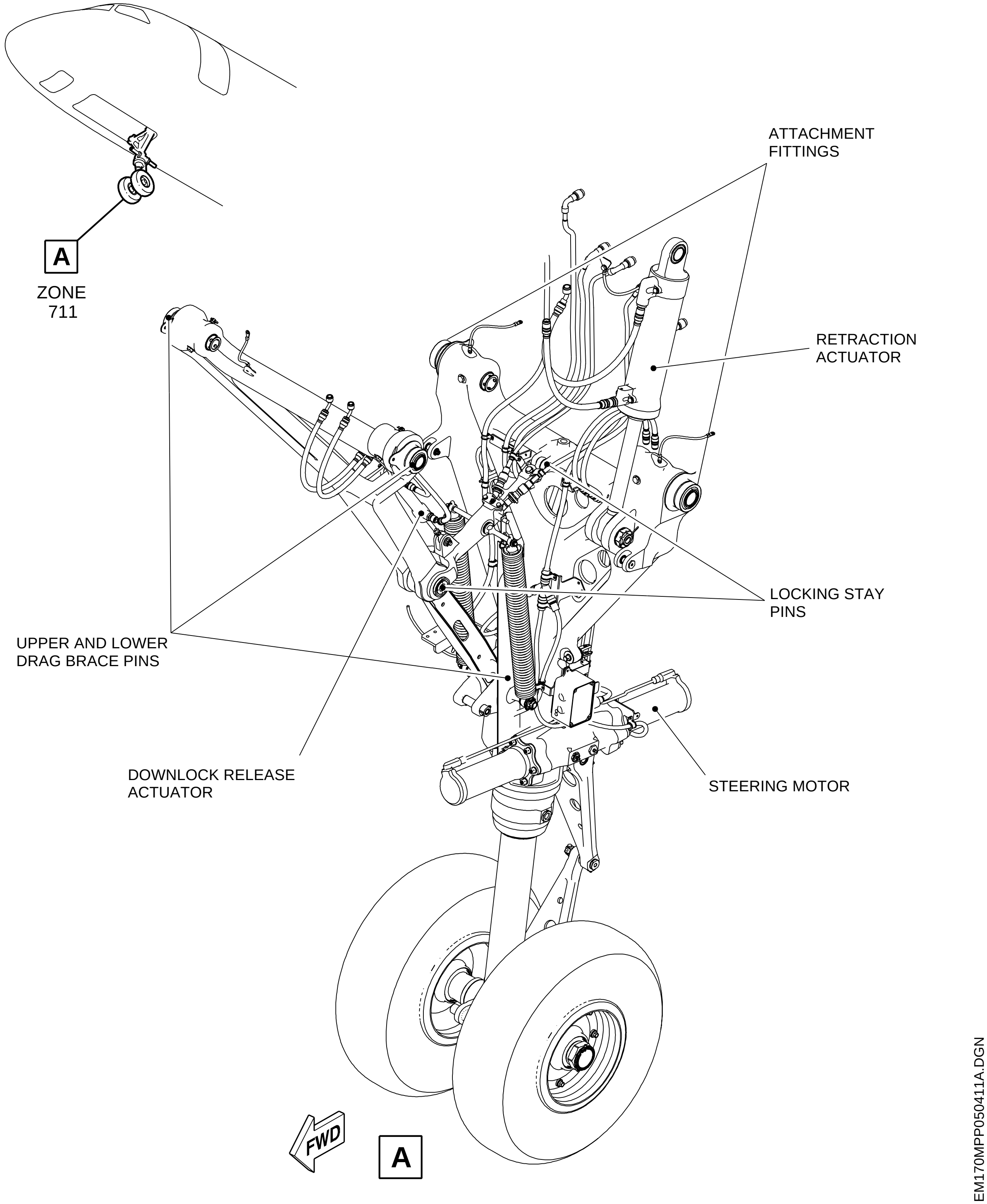
NLG Pins Identification and Structures

Figure 612



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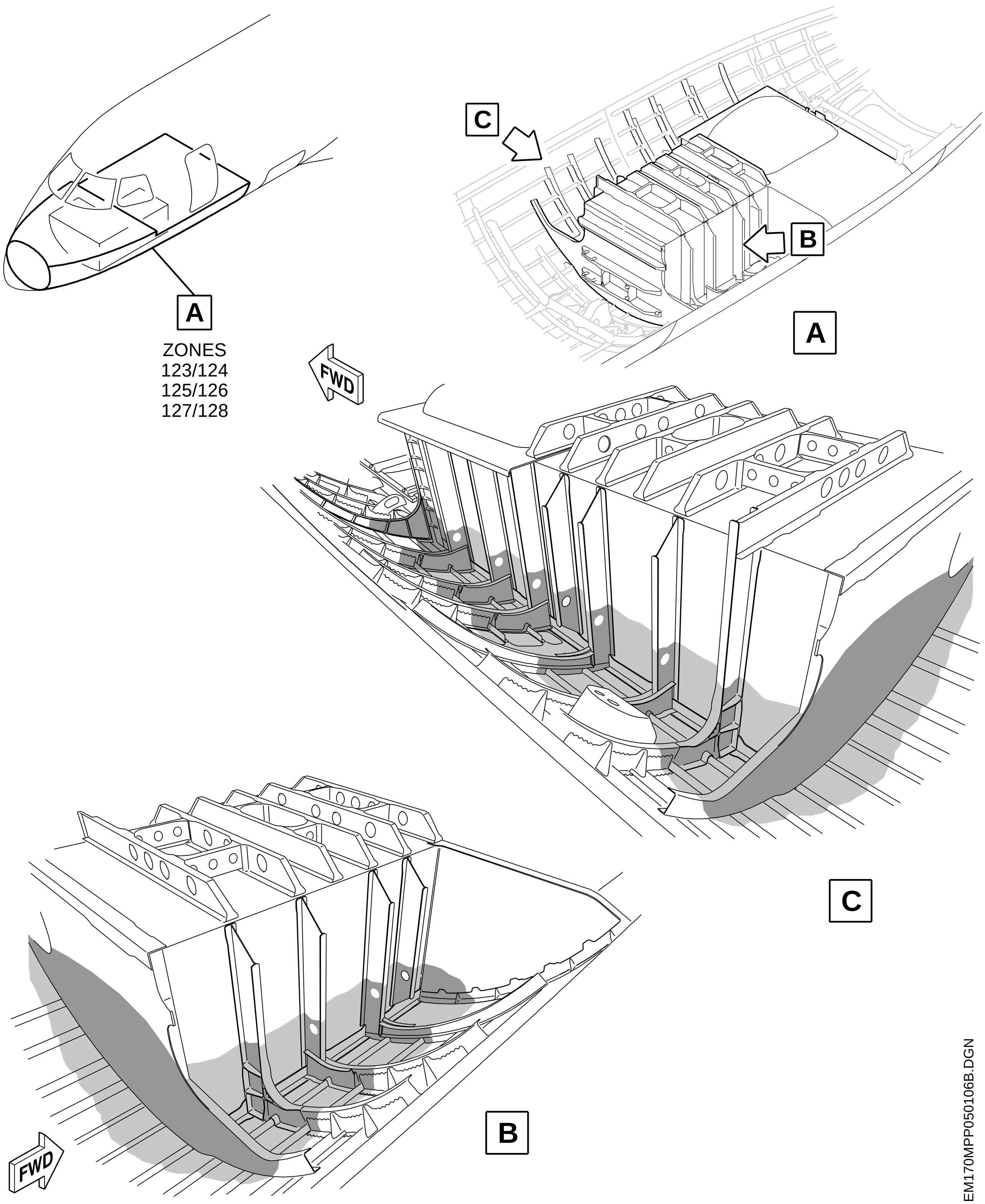
EFFECTIVITY: ALL
NLG - Phase-II Inspection
Figure 613



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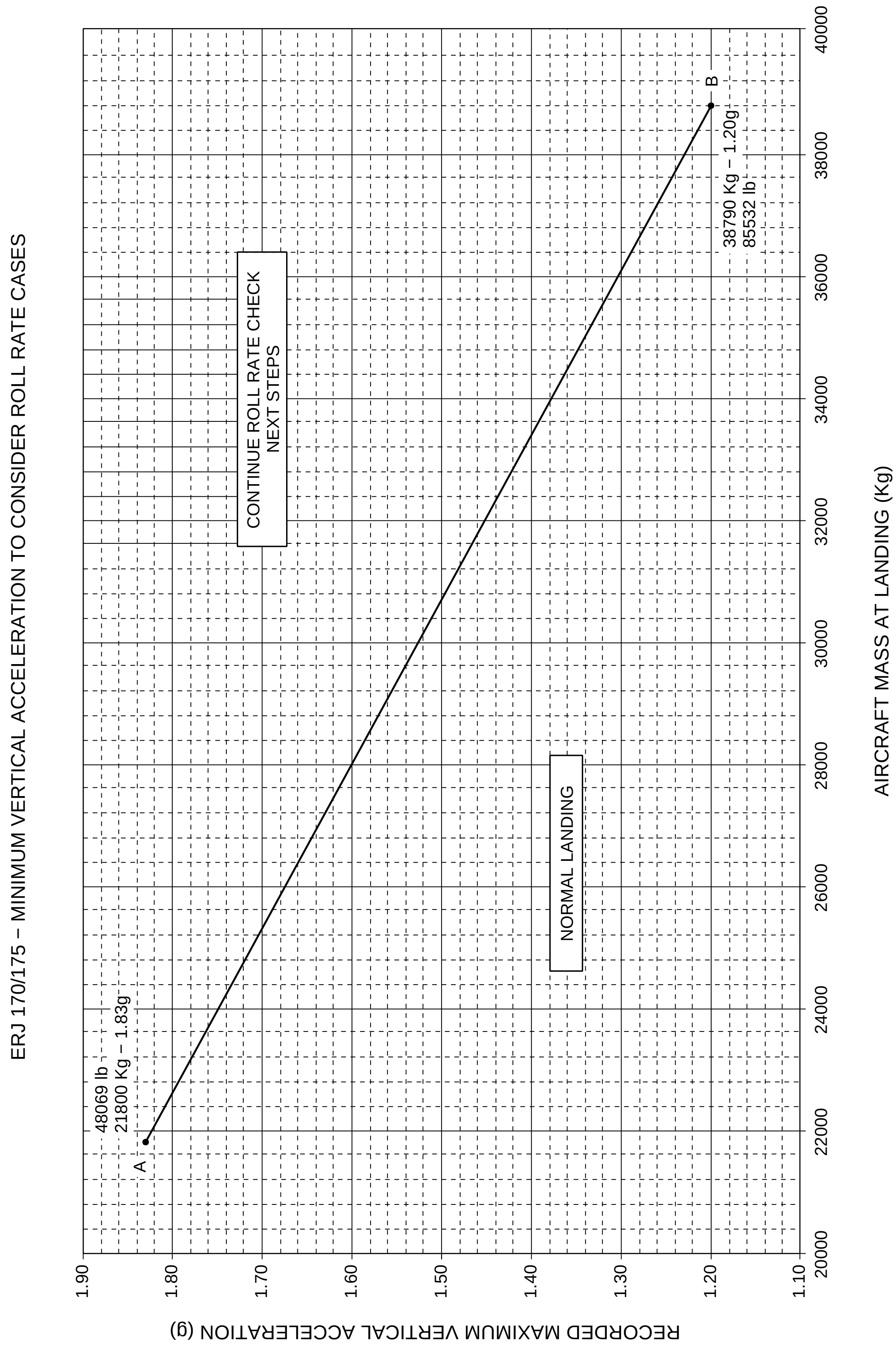
EFFECTIVITY: ALL

Roll Rate Monitor Applicability

Figure 614



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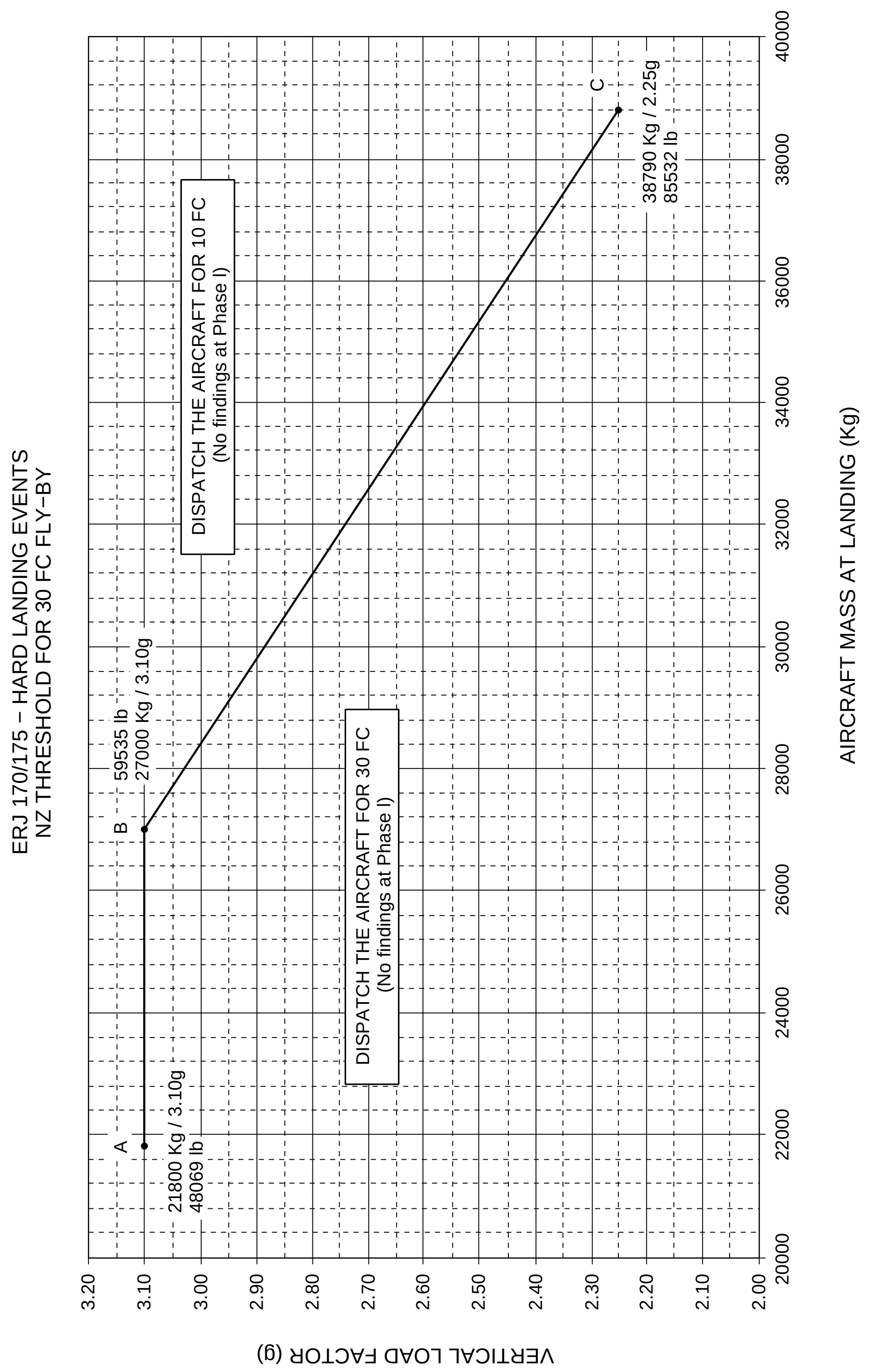
EFFECTIVITY: ALL

Nz Threshold for 30 FC Fly-by

Figure 615



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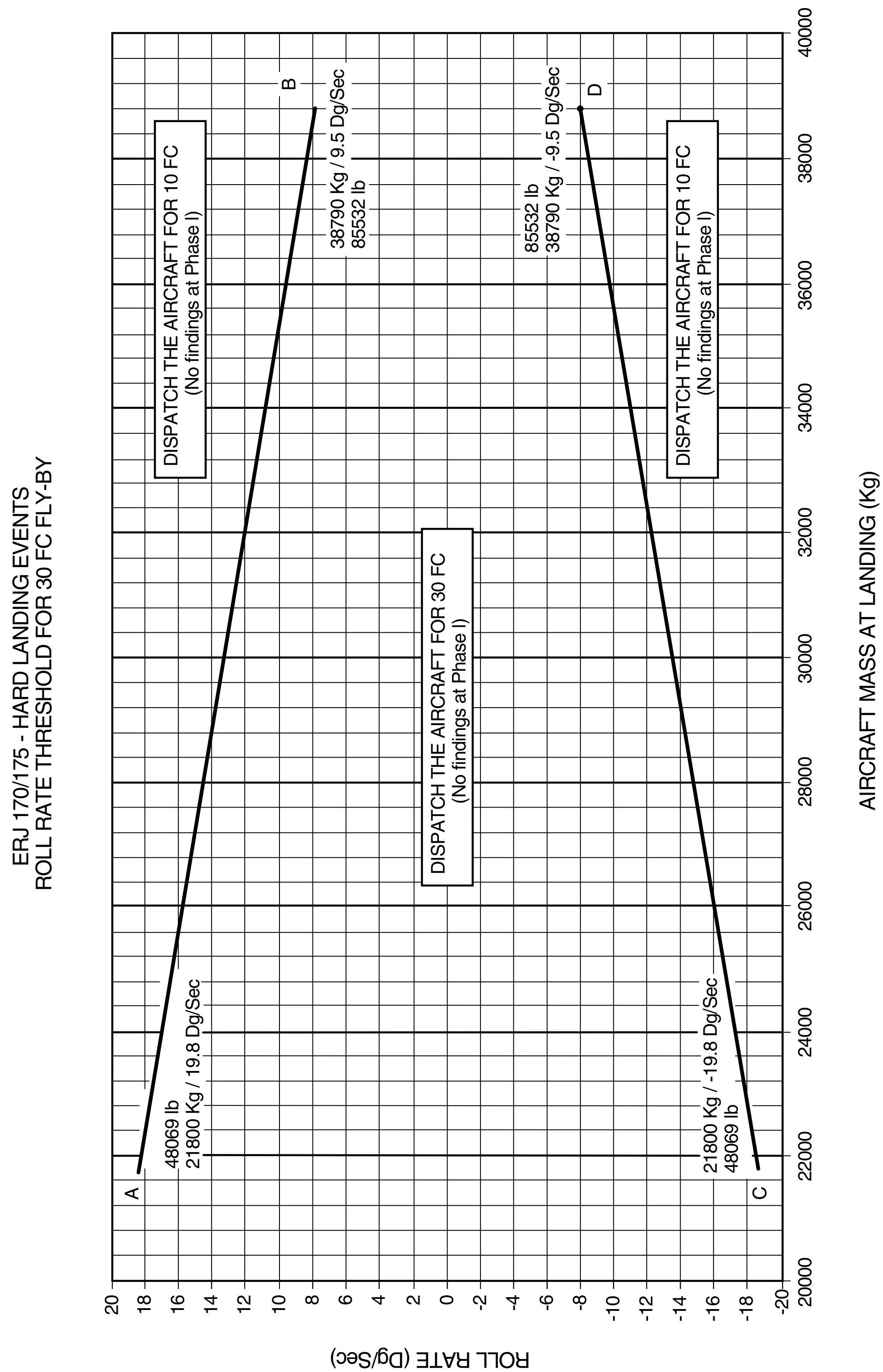
EFFECTIVITY: ALL

Roll Rate Threshold for 30 FC Fly-by

Figure 616



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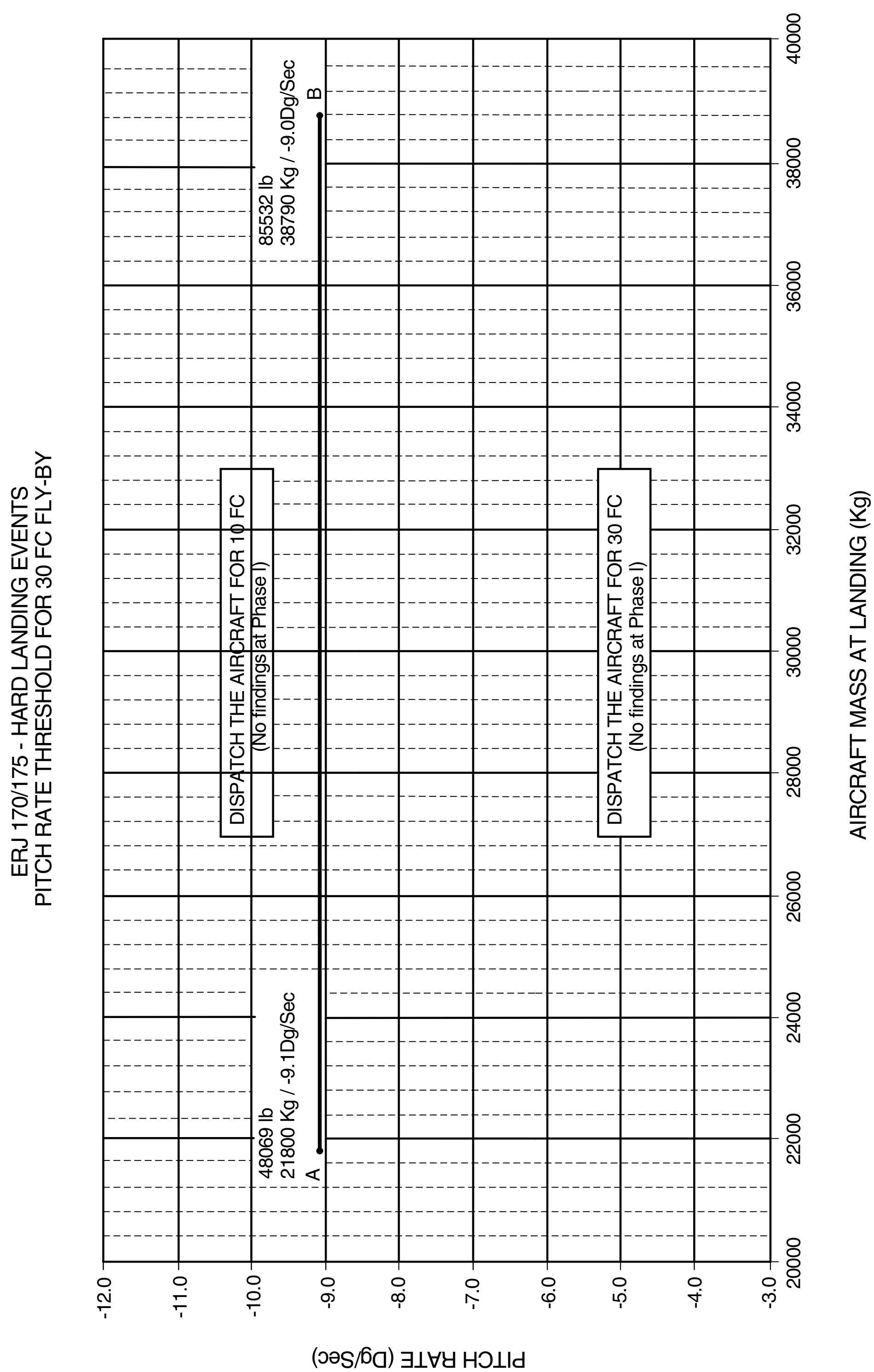
Pitch Rate Threshold for 30 FC Fly-by

Figure 617



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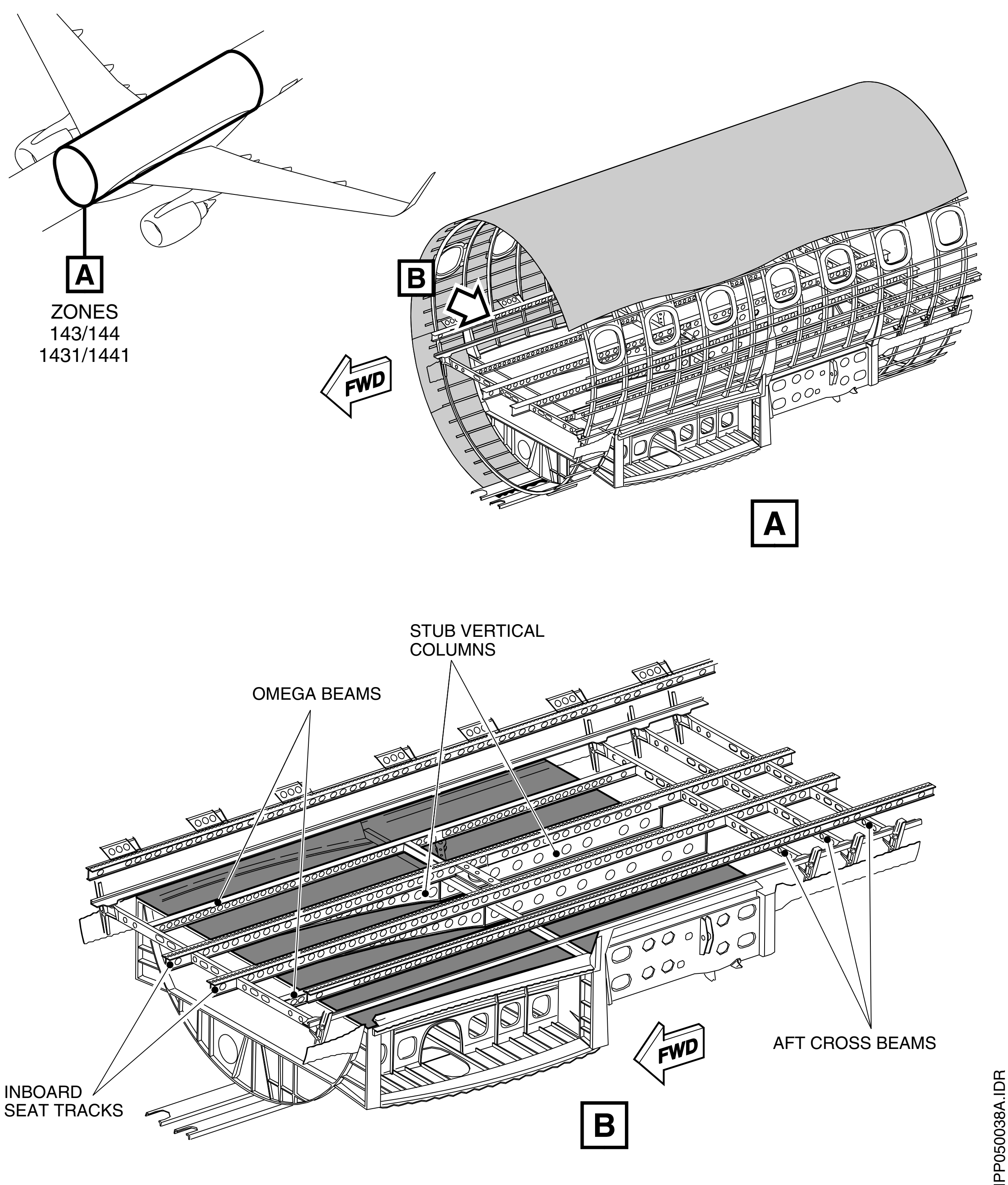
EFFECTIVITY: ALL
CFII-GVI Inspection
Figure 618



LEGEND:
INSPECTION AREA

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